

Predictive Analysis of Renewable Energy production in Wales

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STATEMENT OF ORIGINALITY

This is to certify that, except where specific reference is made, the work described in this project is the result of the investigation carried out by the student, and that neither this project nor any part of it has been presented, or is currently being submitted in candidature for any award other than in part for the MSc Data Science degree of the University of South Wales.

Signed **Md Naser Uddin Sagar** (student)
Date **30-09-2024**

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I would like to take this opportunity to thank my supervisor, Robert Whittey, for his guidance in helping me complete my dissertation.

Abstract:

Moving towards renewable sources of energy is important in combating climate change as well as in establishing long-term energy security. This project aims to address the nature and trends of energy usage in Wales with particular emphasis on forecasting the energy consumption requirements of distinct forms of energy. The first highlighted concern is the imbalance of energy generation and capacity of various renewable technologies, coupled with knowledge gaps regarding global renewable energy trends in relation to the Wales region. To address these concerns, the following steps were followed: data gathering and cleaning for renewable energy, EDA and data visualization for the collected data, employing the ARIMA model for the future energy demands as and for the period of 3-5 years. From the observations, different renewable technologies in terms of electrical capacity and generation with solar electricity gradually increasing while wind electricity remains stationary for Wales. Additionally, trend analysis on Wales data for future forecasting of electrical capacity and generation show positive relation but till 2027, capacity and generation going down. Compared to Global Renewable Energy Production (GREP) data that show almost similar trends for next three years. Further, the correlation analysis reveals low-energy values, which mean that the local factors are more influential in determining the energy status of Wales than the global tendencies. In sum, this project contributes to the understanding of the functioning Wales' power sector and can serve as the basis for further studies of the mechanisms for improving energy sustainability and effectiveness.

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Abbreviations

AI – Artificial Intelligence

DL – Deep Learning

ML – Machine Learning

RE – Renewable Energy

ARIMA – Autoregressive Integrated Moving Average

AIC - Akaike criteria (AIC, Akaike information criterion)

BIC - Bayesian criteria (Bayesian information criterion)

ACF – Autocorrelation Function

PACF – Partial Autocorrelation Function

1. Introduction

The steady increase in the generation of electricity and the more rapid increase in the consumption of electricity has impacted virtually every segment of the global economy creating more reliance on data analysis [1]. From healthcare and finance to agriculture and electricity, the utility of machine learning (ML) and artificial intelligence (AI) has dramatically transformed how industries operate and solve complicated issues [2]. Machine learning, particularly, has emerged as a dominant area that allows for the analysis and interpretation of huge datasets, yielding precious insights and allowing predictive models to gain insights into big data to enhance the system [3]. This transformation is particularly vital for industries that manipulate large portions of fluctuating statistics, consisting of the energy quarter, in which accurate predictions can substantially impact ordinary overall performance, reliability, and sustainability [4].

Globalization in the use of efficient power and enhancing climate change mitigation has extended the conversion of fossil fuels to renewable power resources such as wind, sunlight, and water [5]. This movement enables accurate and certain energy capacity forecasts. Most energy constructions are progressive, and renewable strength assets differ in consistency from conventional resources such as coal or natural gas. As a result, optimizing strength systems and ensuring grid balance require sophisticated predictive models able to deal with the variability of renewable strength inputs [6]. Machine learning has now proved to be an optimal solution for these challenges, one that uses historical and environmental data to predict the capability of electricity from renewable energy systems. These models can be created to learn about the patterns within the data and subsequently reform their prediction to make the energy forecast more accurate and reliable [7].

This thesis explores the application of machine analyzing algorithms in predicting electric-powered capability, focusing mainly on renewable strength systems. via developing data-driven strategy and correct predictive models, this study's goal is to make contributions to greater green energy control, in the long run supporting the global transition to a sustainable electricity future.

1.1. Background

Electricity is the backbone of present-day civilization, powering industries, homes, and transportation systems. However, the environmental effect of fossil fuels, which have historically ruled international energy manufacturing, has ended up becoming increasingly obvious. Greenhouse gas emissions from burning coal, oil, and natural gas have contributed to weather exchange, primarily to extreme weather occasions, growing sea stages, and biodiversity loss. In reaction to those stressful situations, many countries have set ambitious targets for lowering carbon emissions and transitioning to renewable energy sources [8].

Renewable power, which includes solar, wind, hydroelectric, and geothermal electricity, offers a cleaner and more sustainable opportunity than fossil fuels. but, unlike conventional strength sources, renewable strength is inherently variable [9]. Solar energy generation depends on daylight, which fluctuates at some point of the day and 12 months, while wind energy manufacturing is prompted by wind speeds which could change hourly. This variability gives a significant task for grid operators and electricity planners, who need to ensure that delivery meets demand always to maintain grid balance and save blackouts [10]. One of the key factors influencing the combination of renewable power into the electricity grid is the capability to correctly expect electric ability. Traditional methods of electrical capacity prediction rely on statistical models which are frequently unable to capture the complexity and non-linear relationships inherent in renewable power systems. Machine learning, with its capability to version complicated styles and relationships, offers a promising opportunity for improving the accuracy of capability predictions [11].

Machine Learning algorithms can be educated on historical information datasets, inclusive of climate styles, geographic locations, and all other necessary parameters to forecast future power capability. This can be done by taking into account a range of variables that affect energy manufacturing, along with temperature, humidity, wind tempo, and sun radiation, and can adapt to adjustments in these variables over time [12, 13]. With more accurate capacity predictions, machine learning models can be used to better coordinate renewable energy sources and the supply/demand of electricity, that way reducing the dependence on back-up power from sources such as fossil fuel reserves and endorsing a cleaner energy scheme [14].

1.2. Problem Statement

The integration of renewable energy into the grid is a complex challenge for power planners and operators because of the volatile and highly fluctuating nature of renewable generation such as wind and solar. Unlike conventional energy resources, renewable energy cannot easily be regulated, lowered, or raised to meet the demand which is why it becomes challenging to achieve a balance in the supply of electrical energy with the demand on the grid. Energy forecasting is central to these challenges and the maximization of renewable energy. Linear and time series models for energy prediction have their limitations, inadequate in modeling the nonlinear relationship between variables namely wind speed, cloud cover, and temperature which have a direct impact on energy production. Machine learning is the best option as it is capable of modeling these essentialities by learning from a vast number of datasets and applying sophisticated algorithms most effectively. However, there are some restrictions such as a lack of data sources particularly for the remote areas where renewable systems are implemented, and admittedly complex machine learning techniques giving obstacles to broad utilization. Solving these challenges is crucially important for increasing the quality of renewable energy prognosis.

1.3. Aim

The main purpose of this project is to use machine learning trend analysis, and data-driven approaches to forecast the electrical capacity, and electrical generation in regions using different time series models.

1.4. Research Objectives

1. To collect and preprocess the large datasets related to different energy sources to predict the increasing demand for energy in Wales.
2. To compare energy trends in the global context with the consumption patterns in Wales to determine factors that define energy change in the region.
3. To conduct the detailed EDA, to gain insights into different energy technologies and their evaluation over the period for Wales data.
4. To employ the time series model ARIMA model to predict the energy demand for the next 3 to 5 years.

5. To create an interactive dashboard using Power-BI for getting hidden insights into datasets relations all in one chart using filtering phenomenon.

1.5. Thesis Outline:

- Chapter 2, discussed the detailed literature review, exploring recent techniques of energy prediction, datasets, and conventional methods.
- Chapter 3, discussed methodologies and detailed steps employed to gain insights into datasets and compare the technologies mastering the future like clean energy sources.
- Chapter 4 discussed the results and discussion section of the thesis and also offered the best bench comparison of this research with previous techniques.
- Chapter 5, gives a conclusion and overall summary of overall research.

2. Literature Review

2.1. Background

Renewable energy has emerged as a key component of promoting sustainable development and protecting the environment, playing a vital role in lowering greenhouse gas emissions and tackling climate change. Accurate forecasting of renewable-energy production is important for effective energy planning and establishing policies as governments globally want to shift from fossil fuels to sustainable sources. Wales gives a unique environment for researching energy production dynamics due to its wide range of renewable energy sources, such as hydro, wind, solar, and biomass [15]. Predictive analysis in this area not only helps in optimizing the distribution of resources but also improves the reliability as well as efficiency of energy systems. This literature review aims to examine the many approaches and models used in the predictive analysis of renewable energy generation, highlighting significant discoveries, challenges, and potential areas for future study that are particular to Wales. The trends of increasing renewable energy through electrical Capacity (Mwe) and Heat Capacity (MWth) are given in Figure 1, for Wales regions. That shows the exponential increase of energy sources for instance, electrical capacity was 1500MW in 2012, and Heat Capacity was around 0MW that increased in recent years from according to report of Energy generation in Wales [16].

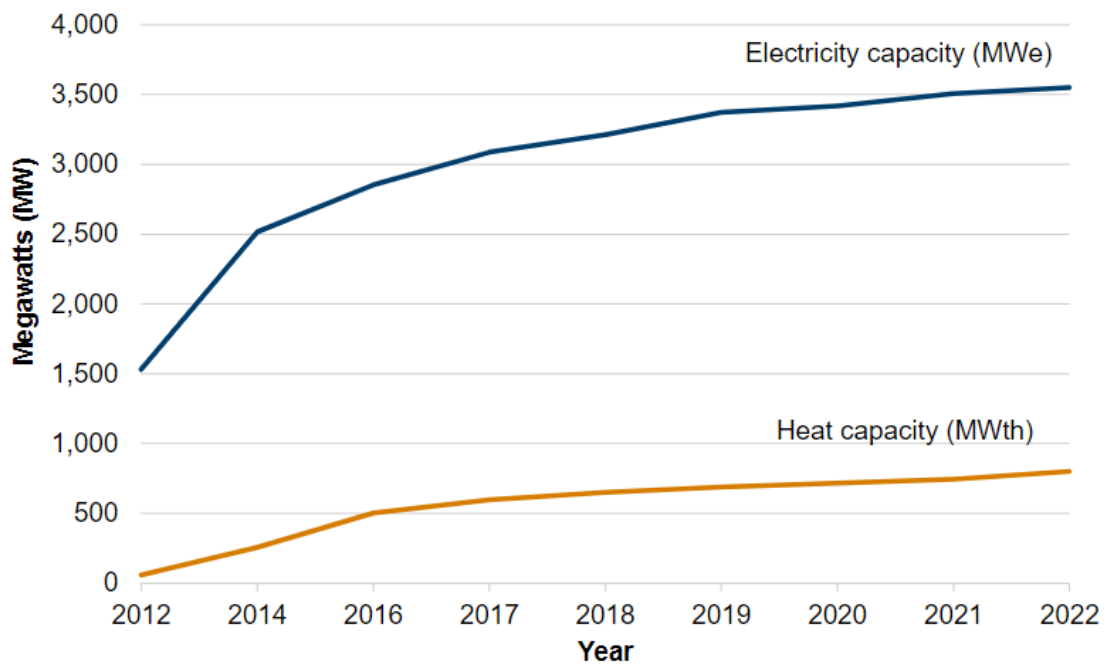


Figure 1: Renewable Energy Production Wales between (2012 – 2022)

According to researchers, most of the work focuses on linear or probabilistic forecasts. It is going to become more important to determine the uncertainty in renewable energy source forecasts [17]. Such projections, known as stochastic forecasts, are more useful than those produced by deterministic models. Several studies have shown how probabilistic renewable-energy forecasts may enhance power system management [18], involvement with the electricity industry and the bidding strategy for ancillary services at Renewable-Energy facilities [19] [20] [21] [22].

For the past few decades, scientists and researchers have investigated sources of Renewable-Energy such as hydroelectric, solar, and wind power. In 2020, Australia produced 24% of its total energy from renewable sources, with 9% coming from solar power, 9% from wind, and just 6% from hydropower [23]. Researchers found that the ocean around Australia is a greater potential source of wave-energy since waves travel great distances with no diminishing their energy in deep water [24], which is also a low-cost alternative to conventional electricity. However, oceanic waves are flexible, unexpected, and influenced by a variety of natural processes. Understanding the associated factors and applying them into a model for forecasting is the key to taking advantage of this energy.

Figure 2 for the world energy consumption from 1990 to 2023, while Figure 3 displays energy source percentages. This time has noticed global energy consumption rise gradually, with

occasional fluctuations owing to economic activity. The western economic crisis in 2009–10 reduced energy consumption growth, and the pandemic in 2020–21 slowed it further. These short reductions just postponed the growth. All fossil fuels are being used more. Oil is still the most extensively utilized fuel, while coal is providing less energy. Both fuels are significant, but the usage of natural gas is growing. Heat, electricity, and chemicals are produced from natural gas (methane). Shale gas extraction, especially in the US, has expanded natural gas supply. Nuclear and hydro power have produced a steady percentage of energy while renewable sources have grown significantly.

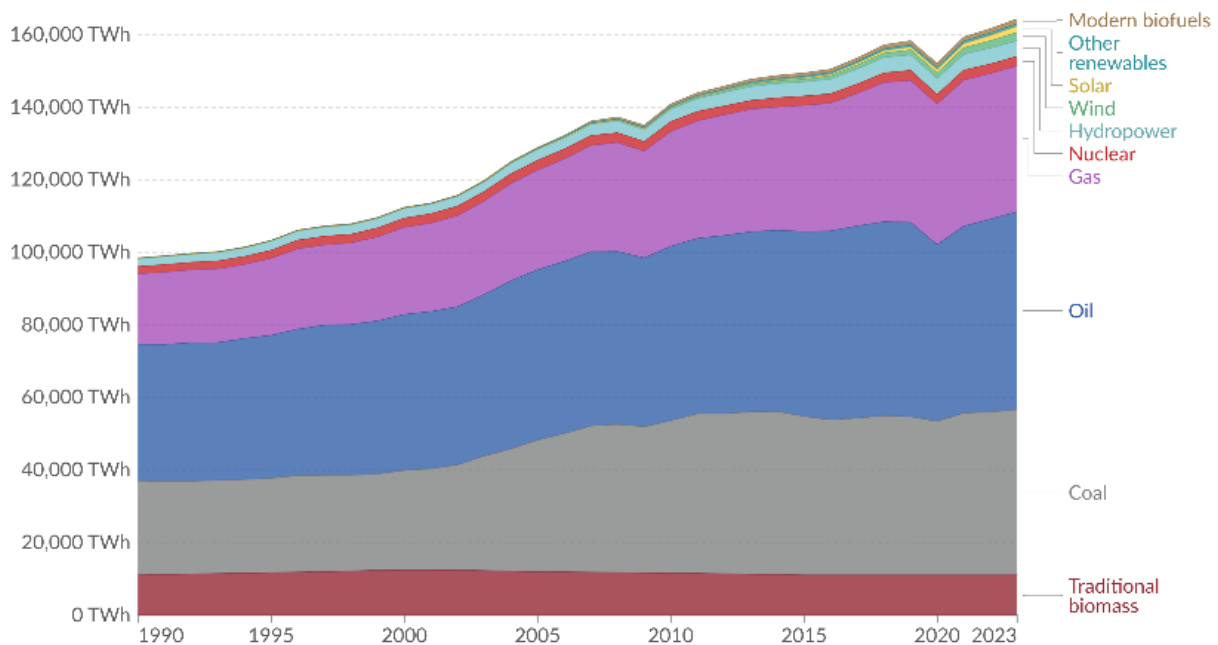


Figure 2: World Primary energy consumption

Source: Energy Institute - Statistical Review of World Energy (2024); Smil (2017)

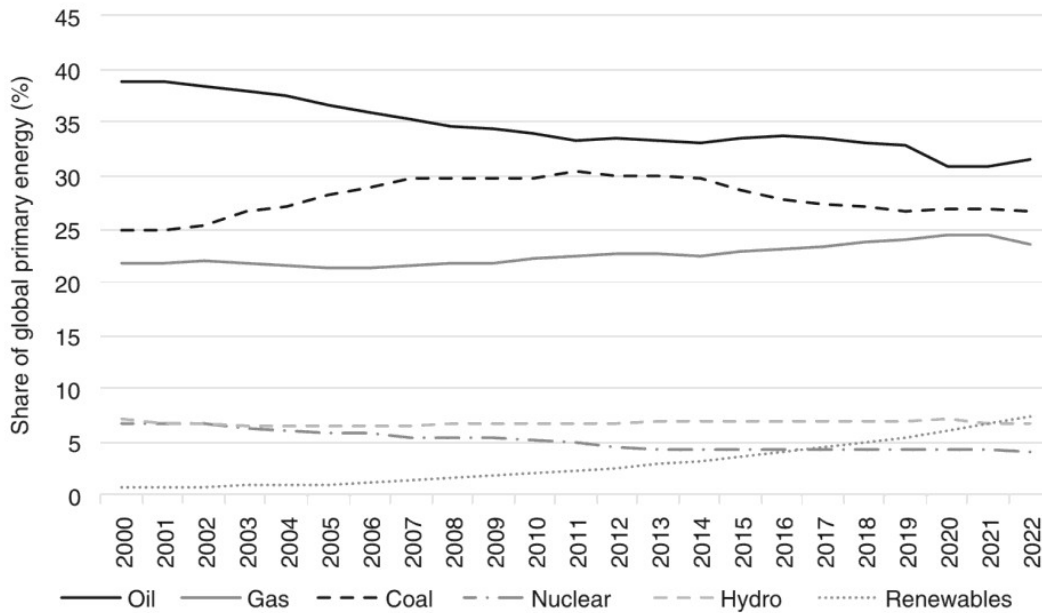


Figure 3: World Share of Primary Energy (Jenkins and Ekanayake, 2024)

Figure 2 shows just economically traded energies rising, which constitutes a minor fraction. It excludes certain traditional sources of wood and other biomass fuels, which are important in some underdeveloped nations for cooking. Wood shortages cause overexploitation, environmental degradation, and excessive costs. Some deprived rural communities face serious problems. Renewable electricity production is rising quickly. Renewable electricity production grew 16% year between 2010 and 2020, as seen in Figure 3 Global solar photovoltaic (PV) producing capacity expanded 38% annually and wind power output 15% annually. These are exceptionally high growth rates, implying that solar and wind energy capacity doubled every two and five years, correspondingly [25].

This paper [26] explores the technological and economic aspects of a rapid energy transition to 2050 using new renewable energy statistics. The research shows that energy efficiency and renewable-energy technologies contribute significantly to this transformation, with synergies. The change is backed by good economics, plenty of resources, scalable technology, and socioeconomic advantages. The results indicate that renewable energy can cover two-thirds of global energy demand and considerably reduce greenhouse gas emissions to keep global temperature rises below 2°C by 2050. The study emphasizes the need for legislative and regulatory frameworks to accelerate renewable energy development six-fold, notably in wind and solar-PV technologies,

and energy efficiency. However, the article highlights that novel solutions, particularly in transport and industry, are needed to eliminate carbon dioxide emissions, which are generally disregarded in international debates. Charging infrastructure and sector-specific consequences should also be addressed.

2.2. Economic and market dynamics of renewable energy

This research [27] analyzes UK solar PV system cost dynamics, power output, and accumulated capacity. Given the UK's aspirations to satisfy expanding demand for energy, fade out fossil fuels, and reach net-zero emissions by 2050, governments, solar entrepreneurs, investors, and the public need the research. The analysis shows that UK solar PV systems' Levelized-Cost of Electricity (LCOE) has decreased significantly during the last decade. LCOE used a large dataset and Two Sets of assumptions to compute historical PV power prices for four system sizes from 2010 to 2021. Based on cumulative capacity and learning rates, the analysis predicts expenses till 2035. Discounted rate, learning rate, project lifespan, load factor, degradation rate, and escalation rate were sensitivity analyzed. Results indicate that large-scale PV systems had a lower middle LCOE of 51 £/MWh (59 €/MWh) in 2021 compared to wholesale energy and other renewable technologies like wind generation. The 0- to 3.99-kW PV system is expected to meet wholesale power pricing by 2027. UK PV systems cost is expected to drop 40%–50% by 2035, making them an affordable power generating choice despite energy storage demands and seasonal fluctuation. Government help, such as streamlined land acquisitions and discounted financing, may boost solar PV system growth, according to the research. Improved subsidies and support programs would help large-scale solar farms expand quickly, easing the transition to a carbon-neutral future [28]. The study finds that solar PV systems are on track to become a significant source of almost free power soon, despite limits and demands for further research on external expenses and seasonal fluctuation.

2.3. Machine Learning Methods in Renewable Energy

Millions of individuals worldwide, including industrialized and emerging nations, are energy poor. Finding energy poverty and employing solutions requires timely and thorough data from surveys, which may be expensive, time-consuming, and problematic to get, especially in rural areas. [29] work utilized ML to identify vulnerable homes in the UK using satellite remote-sensing and socioeconomic survey data. This study provides a machine learning-based strategy to forecast UK energy poverty using the low-income low energy-efficiency (LILEE) signal from remote sensing

and socioeconomic data. Geospatial satellite imagery is combined with local authority data on energy usage, building features, family-income, and other factors. The results show that a ML method using geographical and environmental data can predict 83% of energy-poor regions. This work adds to energy poverty prediction research and may inform UK and global energy efficiency and social equity policies.

Demand response methods for emissions reduction required accurate regional power system carbon emission forecasts. The overwhelming majority of research on short-term forecasting involves recurrent neural networks, particularly LSTM cells. [30] study proposed a PSO-ERT regressor-model for day-ahead emissions intensity forecasting in the Australian-National Electricity market place. PSO-ERT is compared to LSTM, ELM, Multi-layer perceptron (MLP), and Decision Tree. Models are trained with 1.5 years of five NEM region weather, demand, and generation data. ERT beats the LSTM and ELM with Root Mean Squared Error (RMSE) of 42.9, 109.8, 165.7, 25.4, and 74.1 for New South Wales, South Australia, Tasmania, Queensland, and Victoria. For all model's, forecasting performance was better across regions with less energy from renewable sources and when energy from renewable sources was a smaller percentage of demand, suggesting that forecasts of weather could enhance performance.

[31] presents a hybrid deep learning model that uses Convolutional-Neural-Network (CNN), LSTM, and Bidirectional-Gated-Recurrent Unit (BiGRU) to forecast significant wave height (Hsig) for wave energy production. The CLSTM-BiGRU model was designed to forecast Hsig across a variety of time periods (30 minutes, 2 hours, 3 hours, and 6 hours). The input data contains historical wave characteristics such as maximum wave elevation, zero-up crossing wave duration, sea surface temperature, peak-energy wave period, and fascinating wave heights. When tested on multiple wave energy locations in Queensland, an Australian state, the CLSTM-BiGRU model surpassed benchmark models in terms of accuracy, when measured by statistical metrics, scatter plots, and error conclusion. The findings demonstrate the CLSTM-BiGRU model's practical use in wave and ocean energy production, tidal observing, and efficient wave energy resource evaluations.

[32] addresses two important research questions: 'Can an AI-based prediction system optimize solar-energy management?' 'How can the AI-based forecasting framework be integrated into the Smart-Grid system to increase grid reliability and efficacy?' The work uses a Long Short-Term-Memory-based Deep Learning model to improve solar power production projections. AI-

supported planning helps power grid managers ensure a reliable energy supply. The success of this approach is evaluated using performance measures such as MAE, RMSE, nMAE, nRMSE, and R². Using a persistent model for comparison. Although prediction accuracy decreases with forecast horizon extension, the suggested AI framework consistently outperforms the persistent model, especially for horizons beyond two hours. [33] shows that AI-based prediction may improve solar energy management and Smart Grid dependability and efficiency.

Global warming remains one of the biggest issues of our day, as industrial and ICT advances increase greenhouse gas (GHG) production and pollution. Decarbonization techniques intend to replace fossil fuels in power networks with renewable energy sources (RES). These techniques need accurate carbon emission forecasts. Spearman-Correlation-Analysis (SCA) and Improved-Shallow-Denoising Autoencoder (ISDAE) for features-engineering and Improved Particle-Swarm Optimization (IPSO)-based DNN forecaster are used to create a novel and efficient power system carbon emission forecasting model [34]. NRMSE, MAE, and MSE are used to assess the model's performance. Simulation studies demonstrate that the proposed model is superior to SVM and MLR-based models with NRMSE values of 0.173, 0.38, and 0.34, respectively. This better performance shows the model's carbon emission predicting accuracy. Real-world statistics from Belgium and the US are used to assess how RES integration affects electricity prices, consumption costs, and GHG emissions. RES integration may help power system decarbonization, cutting emissions and economic costs, according to the report. The paper emphasizes the strategic relevance of precise carbon emissions forecasting and RES measurement for future decarbonization efforts via descriptive, numerical, and graphical analysis. The model will be applied to additional RES datasets, such as solar also hydro power, alongside South Asian regions like Pakistan. Multiple case studies will measure RES integration's effect in different areas.

2.4. Solar Energy prediction and Optimization

[35] addresses the problem of estimating photovoltaic (PV) power generation in mountainous areas with varying sun radiation. The Radial Basis Neural Network (RBNN) outperforms other data analytics models in forecasting PV power at various intervals (10 seconds to one day), with lowest Root Mean Square Error (RMSE) reported for a wide range of intervals. Specifically, RBNN had RMSEs of 19.11 W (10 seconds), 22.83 W (1 minute), 25.94 W (30 minutes), and 9.08 W (1 day). The Kernel-Ridge-Regression (KRR) performed best throughout the 1-hour interval, amongst an RMSE of 23.22 W. The study shows that including simple attributes such as irradiance and

temperature can improve the accuracy of PV power predictions, offering advantageous insights for improving PV system efficiency and energy planning in a variety of weather conditions.

[36] study compares EPC-modeled energy use to smart-meter data collected for 1,374 gas heated British households and finds significant variances. EPCs typically overestimate energy use, with gaps ranging from -26 kWh/yr/m² (-8%) in the C band to -276 kWh/yr/m² (-48%) in bands F & G. Additionally, the difference persists even when homes match EPC estimates for occupancy and heating, implying that additional factors must affect. This research highlights EPCs' weaknesses in properly estimating energy use, underlining the need for better models. Addressing the highlighted deficiencies could enhance the reliability of EPCs and foster more effective energy-efficiency plans.

In recent years, modern information technologies such as DL and big data have been utilized to enhance energy efficiency in the development of energy management systems. [37] research used a RNN to forecast photovoltaic (PV) generated electricity using previously gathered meteorological and PV power data. The study included a variety of optimization methods, such as normalization, learning data classification, and layer option choice. The accuracy of the model was determined by a coefficient of variation in the root mean squared error ($Cv(RMSE)$), which yielded 13.8% for a single-layer version and 13.2% for a multiple-layer model. Although no significant differences were found between the two models, the multiple-layer model had a somewhat lower error rate. The results indicate that an accurate prediction of PV power-production is possible, revealing the way for a stable energy distribution system via energy storage management and urban-scale energy demand forecasting.

Grid operators face significant demand forecasting issues as distribution feeders include variable renewable technologies, particularly solar PV plants. Changing after load to Net-load-forecasting (NLF) is necessary. This research [38] proposes a Bayesian Neural-Network model for direct short-term NLF at the distribution level. Decision heuristics based on statistical post-processing (clustering of daily-irradiance-patterns) increased the suggested model's performance. Model validation was completed with historical statistical weather forecasts and net load data from 3-Cyprus distribution feeders (2.5%–34.2% PV percentages). The improved model has excellent predicting accuracy, with $nRMSE < 5\%$ for distribution feeders. Statistical post-processing enhanced the model's predicting accuracy, obtaining $nRMSE < 1.3\%$. Finally, the NLF technique

was shown to work for distributed feeders with various PV penetration shares, making it useful for distribution system operators' decision-making and planning.

2.5. Wind Energy Prediction and Optimization

Quantiles for regional renewable energy output forecasts in Spain, compared to SVQR, GBQR, NGB, and RFs. QRDNN frequently outperformed other algorithms in both the wind (Granada, Lugo) and solar energy (Ciudad Real, Córdoba) areas [39]. It showed improved quantile loss, CRPS, and prediction intervals (PIs), with intervals that were 8-29% shorter than those obtained by QRF. The research also proposed a conformed version of QRDNN (CQRDNN), which improved the interval width while maintaining enough coverage. Overall, QRDNNs, especially in their conformalizer version, excelled in precisely forecasting renewable energy generation and providing exact prediction ranges.

The rapid depletion of non-renewable energy resources and its negative environmental impacts, such as air pollution, need a transition to renewable energy sources [40]. Recently, renewable energy generation has gained assistance from investors, governments, and industries worldwide. The global deployed wind power capacity has grown by 345.24% in the last decade. As wind turbine power outputs increase, robust generation management systems are needed to predict them using actual-time wind velocities. Researchers [41] compared five robust regression models to anticipate wind power generated by turbines using wind velocity vector components. The extreme gradient boosted regression model yielded the best results. The model's predicted output can be compared to a city's daily power needs to determine whether to shut down turbines to reduce waste or use alternative energy sources to make up for production shortfalls on less windy days.

The need for electricity is rising daily and energy from fossil fuels contaminates the environment. Clean renewable energy sources like wind and solar power are becoming realistic options for meeting electrical energy demand [42]. But solar and wind electricity is intermittent and varies in magnitude. Forecasting both wind and solar power output is essential for renewable energy system planning and management. This research by [43] seeks a model using machine learning for wind power time-series prediction and examines popular machine learning and deep learning methods. This work uses machine learning and historical wind-generated energy production data to estimate power generating output per hour for addressing a renewable energy problem. Thus, historical wind statistics with input features including wind speed, air temperature, wind direction, pressure, and wind power production output parameters are used for both training and testing employing

mean absolute error to evaluate forecast accuracy. The baseline, linear, dense, multi-dense, CNN, and long-short-term memory layers undergo training and evaluation for prediction accuracy. Further investigation shows that LSTM and Residual LSTM have superior mean absolute prediction accuracy of 0.0987 and 0.0958. This illuminates wind power forecasting methods and opens novel research opportunities.

2.6. Statistical and Analytical Approaches in Renewable Energy

This paper [44] presents a supervised approach for feature-transformation and reduction (SLMVP) to enhance solar radiation forecasts while addressing the limitations of PCA, a popular method for dimensionality reduction in renewable energy. SLMVP, along with PCA, LOL, LPP, and SNMF, have been evaluated on radiation data collected from Seville and Lisbon using linear and nonlinear regression approaches. Research results revealed that SLMVP outperformed other approaches in terms of MAE when considering GHI and DNI radiation at both locations. The advancement of SLMVP above other approaches varied between 1.73% and 6.24%. The study suggests that SLMVP is a robust and competitive approach for reducing dimensionality in solar radiation forecasting, offering potential advantages for different machine learning algorithms.

By the end of March 2019, the UK Government removed the Feed-in Tariff (FiT), which subsidized low-carbon power producers. This has raised concerns about small scale production in the energy grid. [45] research analyzes the variables that affect the adoption of solar photovoltaics in England and Wales and its regional distribution to help navigate post-subsidy unpredictability. Research focuses on social, housing, political, energy, and environmental issues utilizing Ofgem FiT installation data at a comprehensive geographic resolution from April 2010 to September 2019. It has been demonstrated that population demographics, housing density, size, type, and tenure, and energy consumption behaviors influence local PV adoption. Controlling for other circumstances, the Southwest and East of England have unusually high uptake. This is the first fine-grained geographical PV uptake model for England and Wales.

This research [46] used Artificial Neural Networks (ANNs) and the Innovative Sen Method to predict short- and long-term energy production for two hydroelectric projects in Turkey's Sakarya Basin. Researchers used data from the Hydroelectric Energy Plant Operation Directorate, which included day-to-day produced energy, daily average net drop, and daily average flow. The ANN model has three-layers: 1st an input layer with 32 neurons, a hidden layer with three sub-layers (24, 12, and 8 neurons each), and an output layer. Both hydroelectric projects had great dependability,

as proven by their R^2 values. The ANN model for Pamukova HEPP demonstrated no significant change in energy output between 2015 and 2030, however the Sen Method showed declining trends in several middle-region parameters. The ANN model forecasted consistent energy output for the Adasu Regulator and HEPP, nevertheless the Sen Method revealed mixed patterns, with rises in the low and high areas and drops in the central region. These findings are important for future energy planning and highlight the value of integrating ANNs with statistical approaches. Renewable Energy consumption by the world is shown in Figure 4.

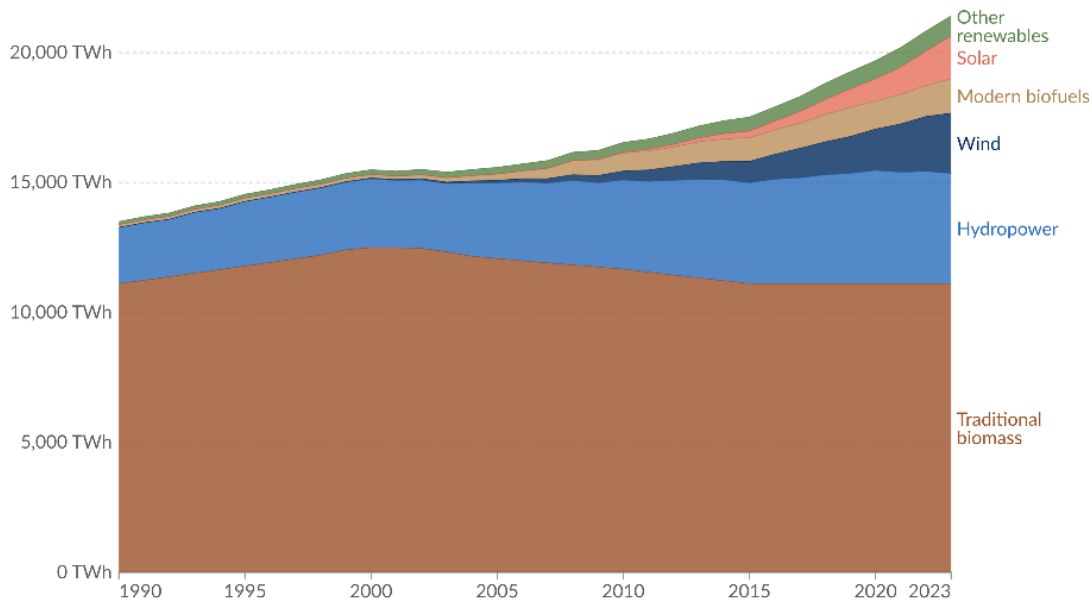


Figure 4: Renewable Energy Consumption in world

Source: Energy Institute - Statistical Review of World Energy (2024); Smil (2017)

Renewable energy sources are needed to address fossil fuels, environmental concerns, substantial transmission and distribution losses, and fluctuating utilization. Weather-dependent renewable energy like solar PV and wind power is most unpredictable [47]. A potential generation forecast can solve this issue. A standard neural-network technique based on ML is suggested for predicting generation and calculating errors [33]. Consumers use power at their discretion; hence a consumer energy management model is needed to produce the projected generation for a self-sustainable microgrid. Consumer energy use must be predicted hourly to create a pricing model. The consumption profile will be calculated using LSTM, which is appropriate for time-series forecasting using sequential-load-Consumption data. Forecasts and analyses are done for fixed, non-shiftable, and shiftable loads. MAE, root mean square error, and evaluation matrix MSE are used to calculate errors. Solar and wind power forecasting RMSEs are 2.472 and 3.034,

respectively. Fixed, non-shiftable, and shiftable forecasting of demand have RMSEs of 0.027, 0.067, and 0.21. The results suggest that the proposed solar, wind, and energy consumption forecasting methodologies are more accurate. The technology may be used to store energy for renewable energy generation and demand control for sustainable energy.

Table 1 Literature Review Summary

Ref	Year	Method	Data	Task	Results
[29]	2024	ML and Satellite Remote sensing	Energy usage, building properties, household income, and other data merged with geographical satellite images	Predict energy poverty in the UK using LILEE indicator	A machine learning technique using geographical and environmental data predicts 83% of regions with significant energy poverty.
[31]	2024	CLSTM-BiGRU	Historical wave characteristics data from Queensland, Australia	Forecast significant wave height	Model surpasses benchmark models in accuracy; practical use in wave energy production
[32]	2024	LSTM based DL Model	Solar Power generation Data	Optimize solar energy management	AI framework improves solar power production projections; outperforms persistent model
[35]	2024	(RBNN) & (KRR)	2.680 kWp PV system, 8 PV modules (335 W each), experimental setup at National Institute of Technology Sikkim	Predict PV power output at various intervals (10 seconds, 1 minute, 30 minutes, 1 hour, 1 day)	RBNN: RMSE = 19.11 W (10s), 22.83 W (1m), 25.94 W (30m), 9.08 W (1d); KRR: RMSE = 23.22 W (1h)
[34]	2023	SCA, ISDAE, IPSO-based DNN forecaster	Real-world data from Belgium and the US	Carbon emission forecasting	Model superior to SVM and MLR; accurate carbon emission predictions
[27]	2023	LCOE, sensitivity analysis	Dataset on UK solar PV systems	Analyze cost dynamics, power output, and accumulated capacity	Significant decrease in LCOE; projected cost reductions by 2035
[30]	2023	PSO-ERT regressor	Weather demand data of 1.5 years from NEM regions	Day-ahead forecasting of emissions intensity for NEM	The ERT model outperforms with Root Mean Squared Errors (RMSE) of 42.9, 109.8, 165.7, 25.4, and 74.1 for the states of New South Wales, the state of Tasmania, South Australia, Queensland, and Victoria, respectively.
[45]	2023	Statistical Analysis	Use Ofgem fit-installation data from 2010-2019	Analyze factors influencing domestic PV uptake in England and Wales	Demographics, housing, and energy use are important variables. SW and EE England have a high uptake. PV adoption is greater among retirees, skilled workers, and detached houses.
[36]	2023	Statistical Analysis, Linear Regression	EPC data, smart-meter energy use data from SERL Observatory	Compare EPC-modelled and smart-meter measured annual energy use	EPCs overestimate energy use; discrepancies increase with lower EPC bands, with gaps of -26 kWh/yr/m ² to -276 kWh/yr/m ² .
[38]	2023	Bayesian Neural-Network model	Historical statistical weather forecasts and net load data from Cyprus	Short-term Net-load-forecasting (NLF)	Improved model with high predicting accuracy; useful for decision-making and planning
[48]	2023	Extreme Gradient Boosting regression	Forecasting wind power output based on wind velocity	Forecasting Wind Power	Performance-evaluation on Testing-data: MAE 0.0524, MAPE 3.283% RMSE 0.1073.
[43]	2023	ML, deep learning methods	Historical wind-generated energy production data	Wind power time-series prediction	LSTM and Residual LSTM superior in prediction accuracy; illuminates wind power forecasting methods
[49]	2023	SLMPV	Solar radiation data from Seville and Lisbon	Dimensionality reduction for solar radiation forecast	PCA, LPP, LOL, and SNMF were all outperformed by SLMVP. MAE improvements were also found in Seville GHI (6.24%), Seville DNI (3.60%), Lisbon GHI (1.73%), and Lisbon DNI (4.50%).
[39]	2023	QRDNNs	Renewable energy production data from Granada, Lugo (wind), Ciudad Real, Córdoba (solar), NWP variables	Forecasting regional renewable energy production	QRDNN outperformed SVQR, GBQR, NGB, and RFs in mean quantile loss, CRPS, and PIs metrics; QRDNN intervals 8-29% narrower than QRF, better coverage and width ratio
[37]	2021	RNN	Weather and PV power generation data,	Predict PV power generation	Cv(RMSE): 13.8% (single layer), 13.2% (multiple layer); multiple-layer model showed slightly reduced error rate

3. Methodology Renewable Energy:

The data set of renewable energy prediction was extracted from the official website of Stats Wales under the catalog for environment and countryside energies section. This dataset named as Low Carbon Energy Generation by Local Authority and Technology gives information regarding the generation of energy through low carbon technologies and authorities in Wales. The information encompasses, the type of technology utilized, the energies produced, as well as geographical splits which are very important especially when analyzing regional energy dynamics. Using this dataset, the analysis can highlight future energy yields, the behavior of renewable electricity, and testing low-carbon generic technologies in specific regions [50]. The data contain different years from 2012 to 2022 layout of the site is given in Figure 5.

Technology (Total)

Year (2022)

Technology

Area Code

Year

Measure

Local Authority

		Number of Projects	Electrical Capacity (MWe)	Heat Capacity (MWth)	Electrical Generation (MWhe)	Heat Generation (MWth)
Wales		95,047	3,551	798	7,852,465	2,547,017
Wales	North Wales	25,973	638	241	1,274,458	897,894
	Mid & South West Wales	(r) 36,831	(r) 1,367	(r) 406	(r) 2,776,406	(r) 1,182,803
	South East Wales	(r) 32,213	(r) 819	(r) 151	(r) 1,579,389	(r) 466,071
	Offshore	3	726	0	2,221,548	0
	Unknown	28	0	0	664	248

Figure 5:Dataset Source

3.1. Wales Dataset Information:

The dataset consists of different numerical columns that describe exceptional factors of energy technology. Key columns encompass:

1. Number of Projects
2. Electrical Capacity (MWe)
3. Heat Capacity (MWth)
4. Electrical Generation (MWhe)
5. Heat Generation (MWth)

The description of each of the given features is discussed below.

3.1.1. Electrical Capacity (MWe)

Electrical capacity is the maximum possible electric power that a power plant can produce at maximum full output. It is essential when evaluating the value that a facility can offer to the electricity grid system. It represents the capacities of the technologies of power generation like windmills, solar panels and hydro power stations. Familiarity with electrical capacity is useful when assessing how much of renewable power can be generated by a particular period and is essential for the energy masterminds to guarantee adequate energy [51].

3.1.2. Heat Capacity (MWth)

Heating capacity is measured in megawatts thermal (MWth) and shows the maximum thermal output a facility will be able to deliver for heating. The determination of this measurement is important for systems that provide heat for domestic, industrial, or district heating. These challenges mirror the effectiveness and design of technologies such as biomass boilers, geothermal plants, and combined heat and power (CHP) systems. Appropriate knowledge about heat capacity is crucial in the energy management aspect, for heating needs and especially in terms of renewables [52].

3.1.3. Electrical Generation (MWhe)

Electrical generation, defined in megawatt hours electrical (MWhe), details the sum total of the electrical power generated by a power station during a particular duration. It is crucial for determination of real energy generation as well as consumption trends. The methods assist in assessing the output of renewable energy systems with the number of installed systems besides predicting the generation capacity in the future from generation in the past. Supervising electrical generation is important for the grid operations as well as for proper matching of supply with demand [53].

3.1.4. Heat Generation (MWth)

Heat generation expressed in MWth represents total thermal energy used for heating in a given period. This metric is especially relevant for systems using renewable sources of energy in supplying heat, including solar-only heated systems and biomass heating. The integrated heat data thus enables the evaluation of heating technologies performance, decision making on policies

regulating heating solutions and right implementation of renewable heating systems to the energy systems [54].

3.2. Wales Data Preprocessing

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2808 entries, 0 to 2807
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Wales Regions                        2808 non-null   object
1   Counties                             2808 non-null   object
2   Number of Projects                  2808 non-null   object
3   Electrical Capacity (MWe)           2808 non-null   object
4   Heat Capacity (MWth)                 2808 non-null   object
5   Electrical Generation (MWhe)         2808 non-null   object
6   Heat Generation (MWth)               2808 non-null   object
7   Technology                           2808 non-null   object
8   Year                                2808 non-null   int64
dtypes: int64(1), object(8)
memory usage: 197.6+ KB
```

Figure 6: Dataset Information

After downloading the dataset given in Figure 6, the next fundamental step is data preprocessing that involves preparing raw data in a learnable format for making clean, consistent, and structured data available to models for further analysis of the same. It starts with data cleaning wherein the removal of errors such as missing values, duplicate values, and outliers is done. Techniques for handling missing data can include imputation wherein values are replaced by statistical estimates and removal, particularly if missing data is highly frequent. Duplicates are removed to avoid duplication in observations, and outliers are controlled using methods such as Z-scores or the interquartile range, so they would not skew the performance of any model.

Now that the data has been cleaned, another step is data transformation. Data transformation is the process of formatting the data into the requirements of various machine learning algorithms. While numerical features may need to be normalized or standardized. Normalization scales the data from 0 to 1, whereas standardization adjusts it from a mean of 0 and a standard deviation of 1, such that features with different scales are not over emphasized in the model. In this respect, the whole preprocessing scheme ensures that the dataset is properly refined and optimized for training, and leads to even better performance and generalizability. The detailed data pre-processing steps discussed below.

3.2.1. Handling Missing Values:

NaN values in a dataset often indicate a problem, especially when they don't make sense in the context of the data. One common reason for the presence of negative values is a high number of missing data points. When large portions of missing values exist in data, filling them using simple methods like mean imputation can lead to unintended consequences.

Imputation by mean assumes that the missing data points follow the same distribution as the observed values. However, if a significant portion of the data is missing, this assumption may not hold. The mean is sensitive to outliers, and in cases where the missing values represent a substantial proportion of the dataset, the calculated mean may be distorted. This can lead to imputed values that are either too high or too low, introducing negative values in situations where only positive values should be present (e.g., age, income, etc.). Handling missing or negative values in dataset is essential because, for training a model that gives high predictive performance with low bias there is need to use proper imputations either, mean, mode, median or others as per requirement.

From the observations, the dataset contains a list of 84 missing values in similar rows given in Figure 7 & 8. As this project used data from 2012 to 2022 so, possibly earlier data does not record properly. So, filled the dataset with 0 to maintain the distribution of numeric columns. So, manually from the overall dataset distribution most of the values were closed to 0, in this way missing values for remaining columns replaced with 0 to maintain columns integrity.

```
Missing values in each column:
Wales Regions          0
Counties               0
Number of Projects     84
Electrical Capacity (MWe) 84
Heat Capacity (MWth)    84
Electrical Generation (MWhe) 84
Heat Generation (MWth)  84
Technology             0
Year                  0
dtype: int64
```

Figure 7: Count of missing values in data

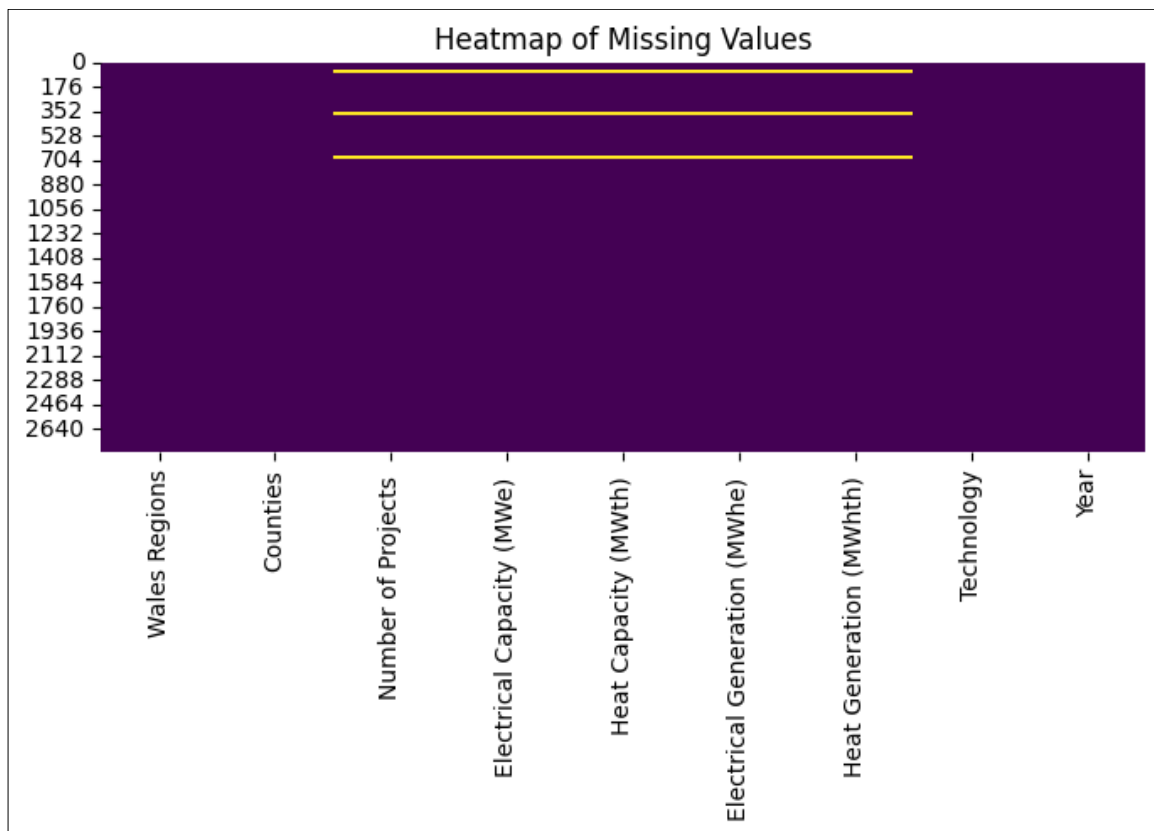


Figure 8: Heatmap for missing values

Furthermore, all columns are numeric but pandas' data frames take them as object so, all these converted to numeric format are easy for further EDA.

```
Wales Regions      object
Counties           object
Number of Projects float64
Electrical Capacity (MWe) float64
Heat Capacity (MWth) float64
Electrical Generation (MWhe) float64
Heat Generation (MWhth) float64
Technology         object
Year              int64
dtype: object
```

Figure 9: Columns Numeric Data Type

3.2.2. Duplicate value:

In this next part duplicate values in the dataset are also checked because, by duplications, if the predictive model is trained with high performance but during deploying time it could deviate from actual results in production. In this case duplicate values in the dataset are 0.

3.2.3. Further EDA:

In this part, after first steps distribution of each column is checked and their distribution for mostly all columns of numeric plotted mention in Figure 10. From the observation, the dataset is high skewed.

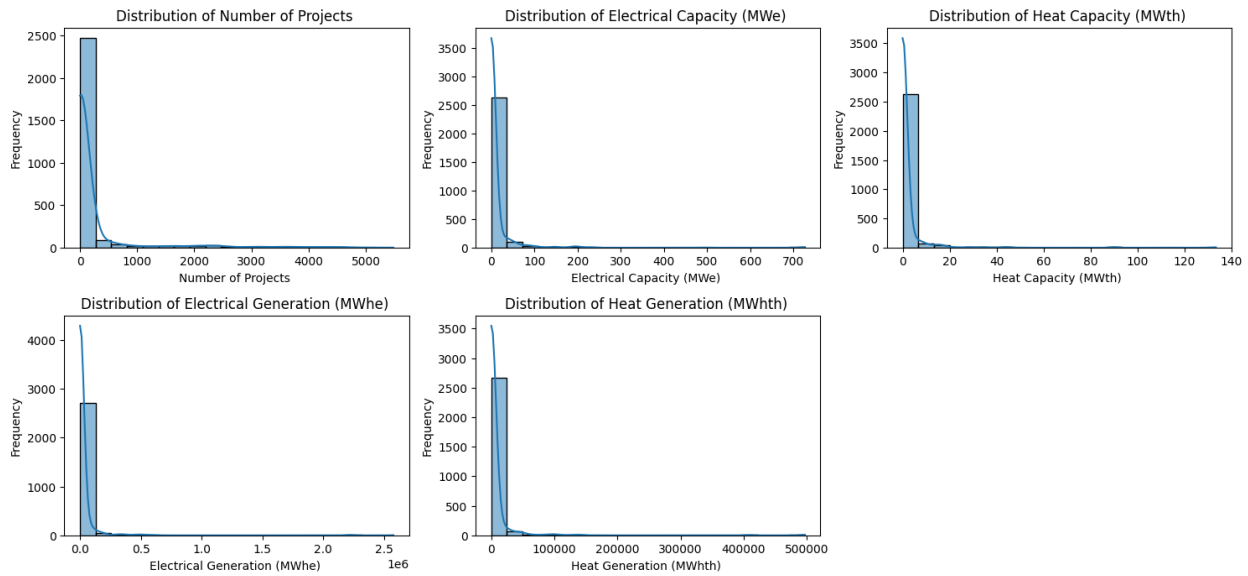


Figure 10: Distribution of Numerical Columns in dataset

3.2.4. Outlier detection and removal Box Plot:

Figure 10, express the columns with right skewed results to further gain into data outlier detection plot box plot is used. A box plot or a whisker plot is a highly standardized method of displaying an array of data and recognizing outliers. It summarizes data through five key statistics: At least, quartile one (Q1), quartile two (the median Q2), quartile three (Q3), and the maximum value are distinguished. The middle large rectangle presents the interquartile range (IQR) that in turn encompasses 50% of the scores, and the lines flooring from the box to each endpoint are known as whiskers that connect to some of the lower and higher values located no more than $1.5(IQR)$ away from the quartiles. If any data value lies outside this range, then those values are declared outliers and are graphed often as points or stars [55]. This method of visual representation is quite

helpful to show both the central measures and scatter measure regarding the average on the same graph; This is vital when carrying out exploratory analyses as it works well in pointing out outliers.

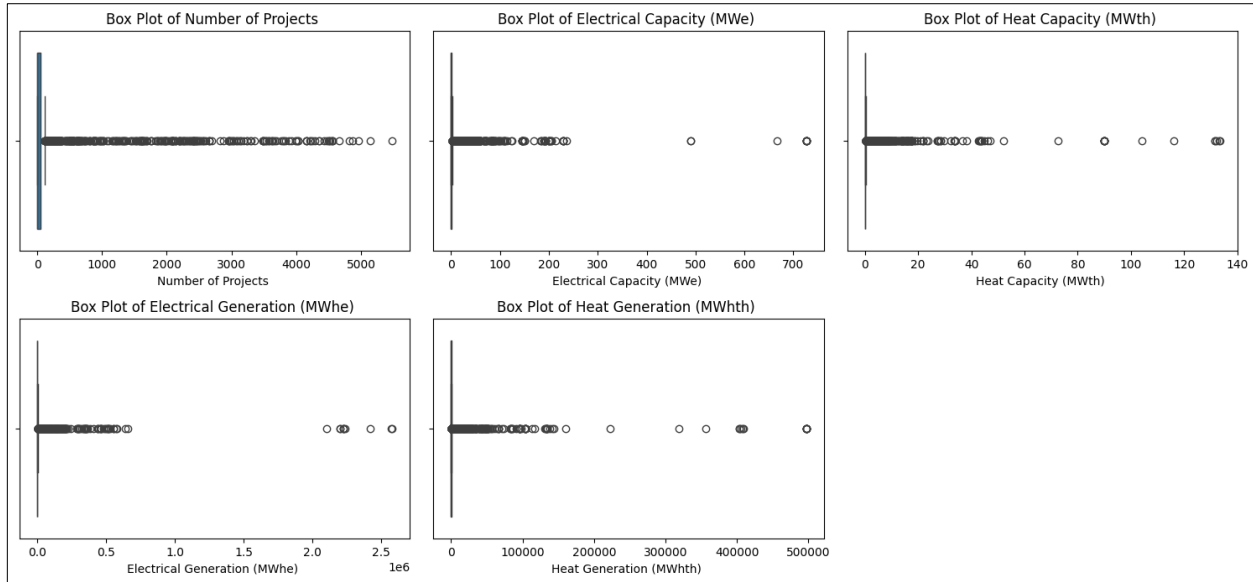


Figure 11: Outlier Detection using Box-Plot

3.2.5. Yeo-Johnson transformation on Wales Dataset

From the above charts given in Figure 10, this can be seen that the dataset is heavily Right Skewed, the maximum peak in the dataset is recorded around 0 that's why there are many more outliers observed in data. In order to deal with such charts there is a need for transformation that can shift the data from right skewed to normal distribution. In this way **Yeo-Johnson transformation** is used. This is a data transformation method that transforms data in an aim of getting an increased uniformity in the outcomes for the purpose of normality. However, it can be applied irrespective of positive or negative values and that makes it suitable for any range type for datasets. It operates through the use of Box Cox transformation in a way that transforms the power of input data depending on the value of the data decreasing the skewness and enhancing the usefulness of data to feed machines learning algorithms. This transformation is impressive especially for methods like linear regressions and others based on random inputs, where inputs are normally distributed. The Yeo-Johnson approach is callable in libraries like Scikit-learn and is proved to optimize accuracy since it attenuates the effect of outliers [56]. The distribution of the dataset after applying the transformation is given below in Figure 12 and 13.

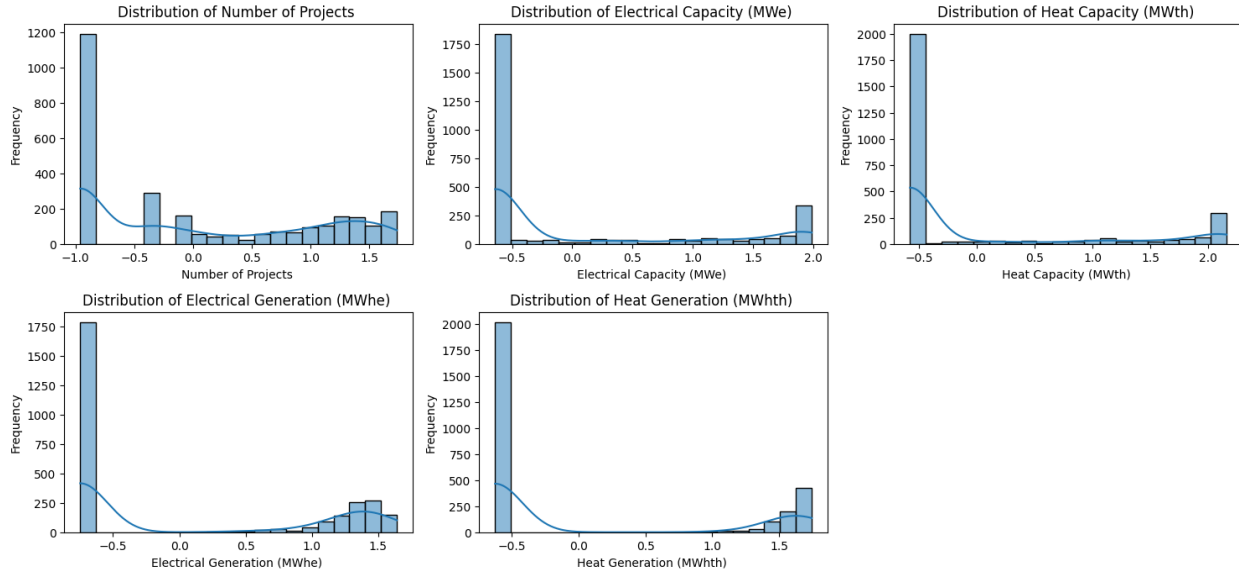


Figure 12: Transformed data after Yeo-Johnson transformation

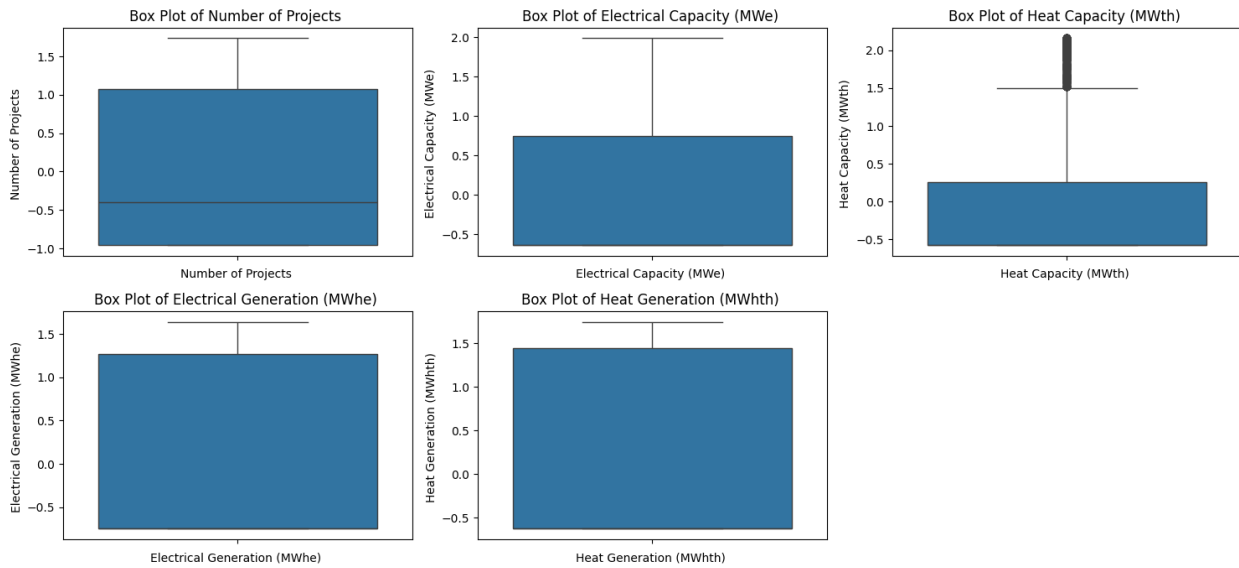


Figure 13: Box Plot after Yeo-Johnson transformation

3.2.6. Technologies vs. Electrical Capacity

In this step correlation is checked between different technologies versus electrical capacity by the technique of visualizing and analyzing the distribution and outliers of electrical capacity for various renewable energy technologies through box plots and distribution plots which would practically help in the better understanding of the capacity variation for various technologies. Several advantages can be found when comparing the electrical capacity of the technologies including Anaerobic Digestion, Biomass, Energy from Waste, Hydropower, Solar PV, Wind

(Offshore and Onshore), and others. It provides the ability to compare the figures of capacity and standard deviation of the different renewable and low carbon generation technologies and assist stakeholders in deciding which technologies offer more capacity factor or stabilized generation capacity. These particularities help the decision-makers to get a clear understanding of the further growth's potential for each technology and the question of its applicability on a regional level. This analysis can be used to inform policy, investment and resource allocation decisions and bring about the optimum use of all forms of energy to bring about a sustainable energy system.

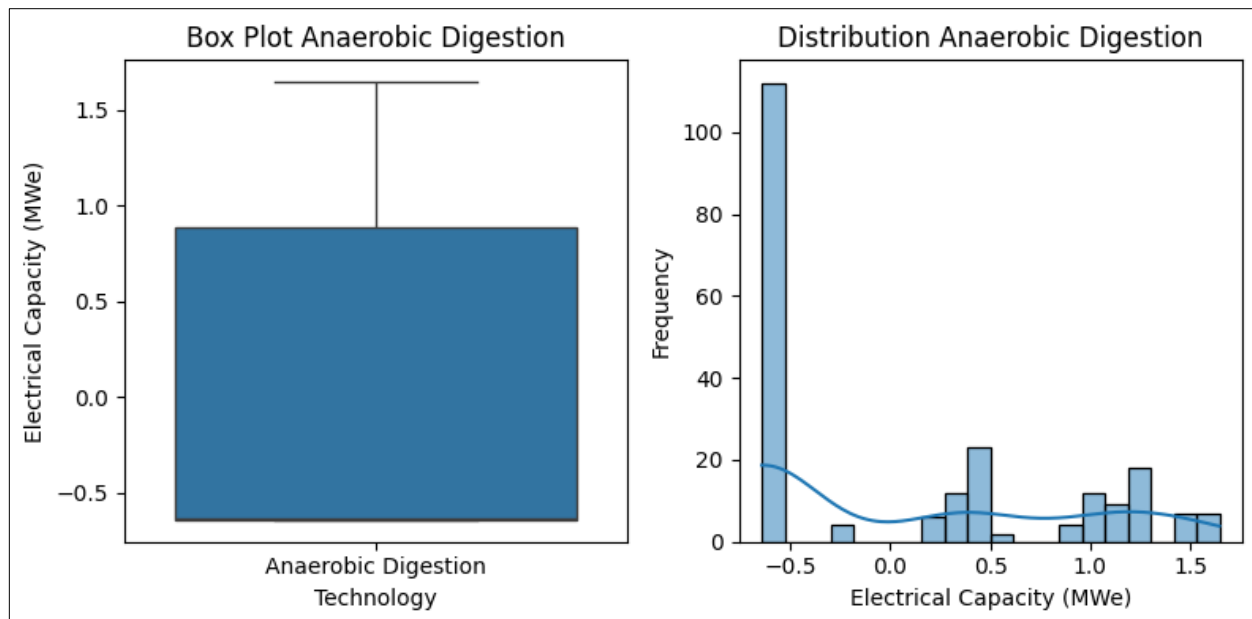


Figure 14: Comparing Technology vs Electrical Capacity

3.2.7. Comparison Between Technologies and Regions

In the same manner further focusing inside the data and patterns to enable comparison between different renewable energy technologies with regards to electrical capacities across the regions in Wales. By using the box plot for every technology and region, the visualization provides clues about regional limitations or strengths with regard to different technologies or capacities of electrical facilities. Division of all plots into individual technologies is useful in case of analyzing isolated regional influences, capacity range, and variation in the performance of each sort of renewable power.

This approach also provides a method for making comparisons to see how each of the various renewable energy technologies work in the various regions across Wales and possibly assist the policy maker, planner or investor in terms of which technologies may suit the area better. Ensuring

that decision makers have a visual sense of relative regional capability to generate energy which can help in directing attention to regions that may need infrastructure investment or development or to regions underutilized for renewable resources. Sample Distribution chart for Technology vs Region given below in Figure 15.

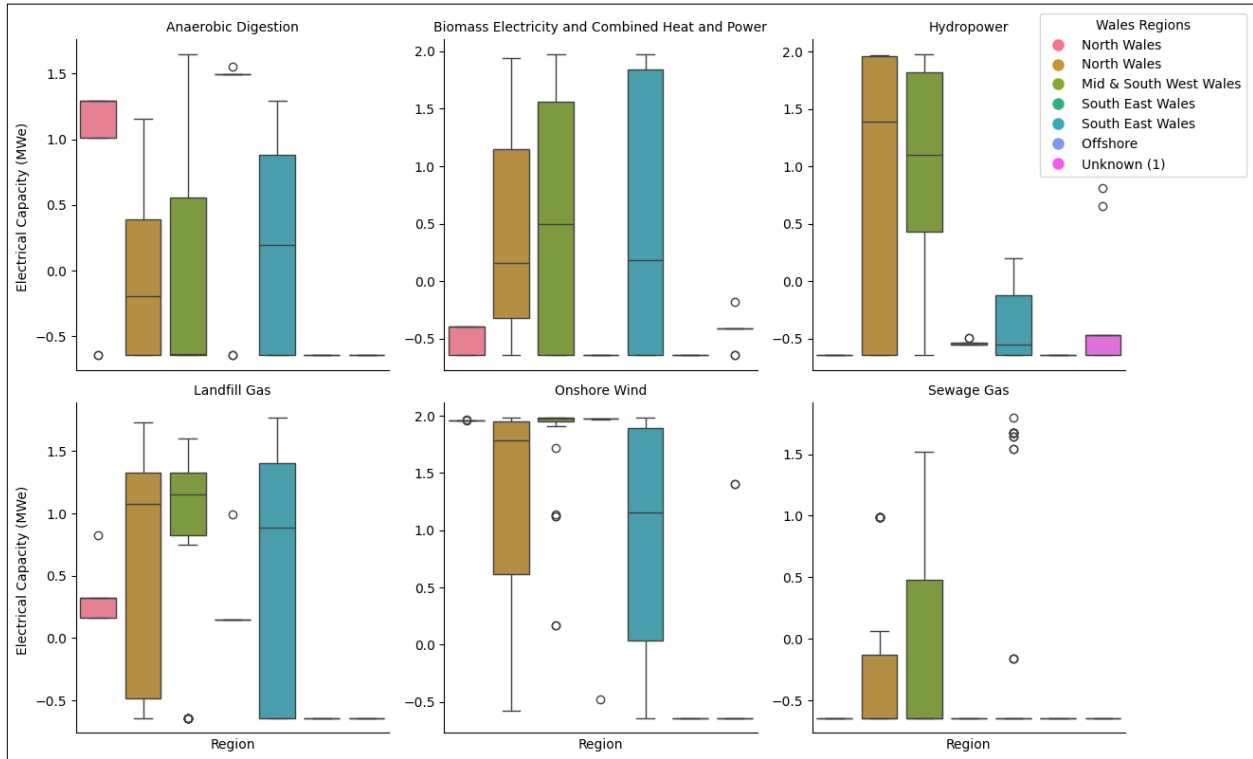


Figure 15: Technology vs. Regions

3.2.8. Correlation Analysis:

Under feature engineering it can be useful to check whether there is a correlation between different forms of energy, which can indicate how these technologies are connected. A correlation plot or heatmap for the purpose of demonstrating the variability or otherwise in the strength of the relationships between different energy sources given in Figure 16. This step makes it possible for one to establish a correlation between the electrical capacity's vs electrical generation or heat generation. When all features are highly correlated, they are simply replicates of each other, and can lead to multicollinearity problems while on the other extreme of having all low correlation or non-correlated features provide the model with more information. With this analysis, feature engineering is enhanced to guarantee the appropriate optimization of the input data for predictive tasks and the improvement of overall model accuracy as well as model interpretability.

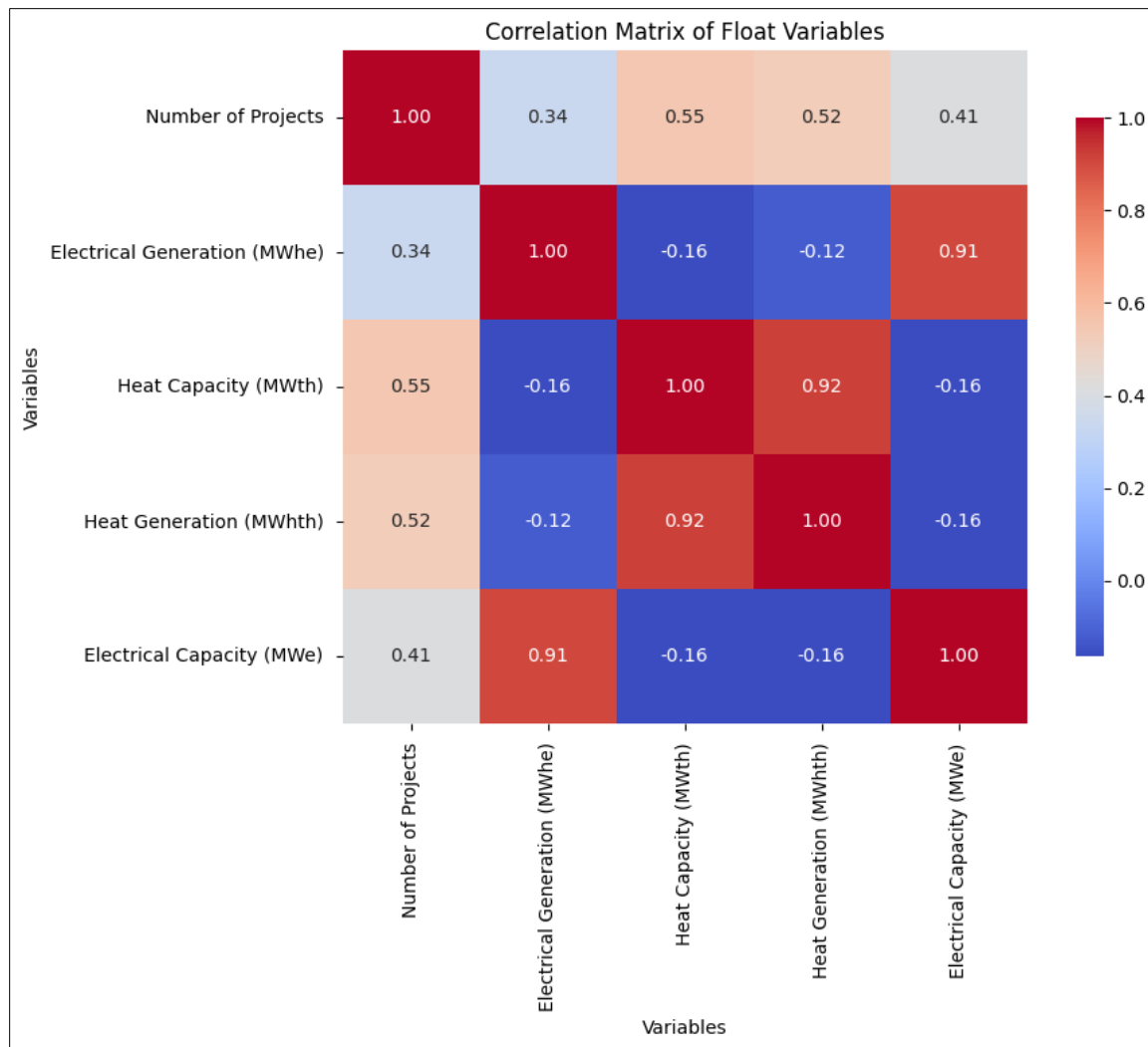


Figure 16: Correlation Analysis

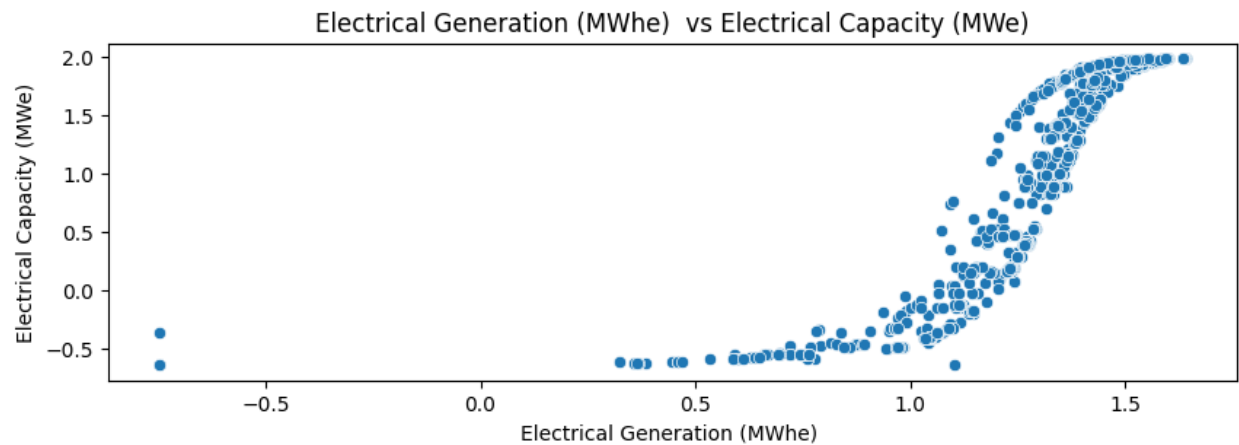


Figure 17: Correlation Analysis between Electrical Generation vs. Capacity

3.2.9. Trend Analysis:

On the method level, the trend analysis is performed to identify time-related characteristics of electrical generation and renewable energy projects capacity given in Figure 18. The data analyzed for the purpose of this report is presented for each year, and the essential indicators include general and capacity electricity generation, general and capacity heat generation, the number of projects, etc. This leads to a clear understanding of how energy production and the capacity has shifted on a yearly basis. The data is then additionally aggregated in a way such that the variation in the electrical power generation and capacity can be represented in a line plot as an overall yearly trend. This approach gives some understanding of increasing or decreasing trends in renewable energy projects and allows analyzing regularities of increasing or decreasing efficiency of energy production and its capacity in the course of its development.

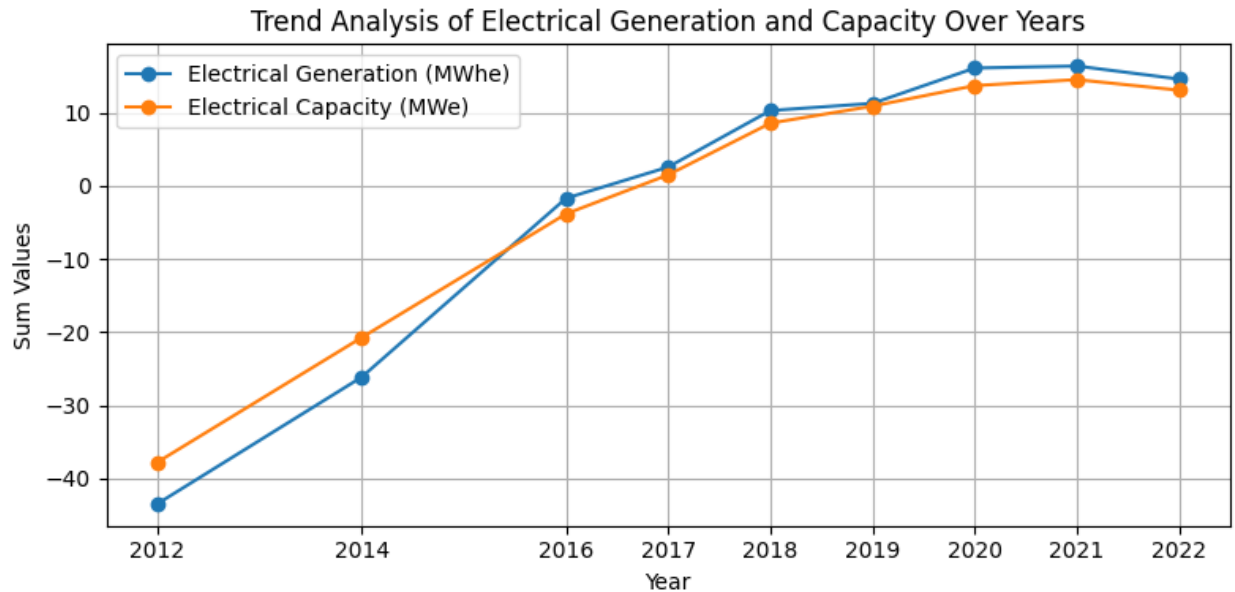


Figure 18: Trend Analysis

3.3. Autoregressive Integrated Moving Average (ARIMA):

For this step, initially, the auto-arma model used to forecast the electrical generation and electrical capacity for the next 5 years. Additionally, after this step, a list of 5 selected technologies including offshore wind, onshore wind, anaerobic Digestion, solar PV and biomass electricity, are used to predict the electrical generation for next 5 years using the auto-arma model.

Auto-Arima function automatically selects the best fitting ARIMA model to the data with consideration to the trends, seasonality and auto correlation of the times series. Finally, after the

auto_arima function from the pmdarima package, the best fitting model for ARIMA was chosen based on the AIC measure. This specific technique got rid of the other traditional approach of choosing from among the various types of ARIMA model. The specified model was subsequently applied to the actual historical electrical generation time series data for each of the renewable energy technologies.

Overall, this clear visualization permits renewal energy generation trends and forecasts to be presented in a way that can be followed further for the analysis and planning. Auto-arima is an algorithm that displays the forecasting of the variable, while historical data characterizes the environment in which the forecasted tendency was formed. Combined, this allows for an assessment of both proportional change and relative interpretation of possible future increases in renewable generation by technology type.

To ensure the assessment of the model selection and the forecasting, ACF and PACF plots are also constructed to determine if there is any correlation of the residuals. The forecast seems believable when viewed against historical data with solar PV maintaining a high growth rate while onshore and offshore wind have more moderate growth rates. The automated ARIMA modeling offers a data driven approach to model and forecast with limited need for model intervention.

By performing technology-specific forecasting, together with various sources of renewable energy can help the investor to develop the infrastructure. The detailed description of ARIMA and Auto-ARIMA models is given below.

ARIMA Model:

The ARIMA model is a rather strong statistical method which is used to make the forecasts for time series. They are particularly useful when it is used to find future values using past values since they capture features such as trends, seasonality and auto-correlation within the data [57].

When it comes to forecasting the generation and capacity of electricity for the coming five years, ARIMA is particularly needed because of the volatility of energy generation and consumption. ARIMA is useful in pointing to cyclical or seasonal factors that may exist or in developing forecasts based on short-term oscillations and long-term trends. Demand forecasting of electrical capacity and generation is important for the management of the grid, energy requirements and future expansion programs [58]. Since electricity generation and capacity has a temporal element in it which can be analyzed using ARIMA due to its modeling capabilities with historical data available it is an ideal choice for making accurate forecasts here.

It works by combining three key components [59, 60]:

- AR (p) is the number of lag observations or autoregressive terms in the model.
- I (d) is the difference in the nonseasonal observations;
- MA (q) is the size of the moving average window.

An ARIMA model order indicated as (AR, I, MA) = (p, d, q) keeping a value to the order or the number of times the function is run in the model. The values of the corresponding parameters of a variable can be zero. In the ARIMA model, different data is used to make the data stationary implying recursiveness of the data at a certain time. This function helps eliminate a trend or seasonality such as market or economic data like in the figure above.

ARIMA is becoming one of the tools popular among the data scientists in order to work on the predictions for future demands, this could be on the sales forecast, manufacturing developing plans or even the stock prices. For instance, when used in forecasting stock prices, the model captures disparities between values in a series more than it can capture the actual values [59, 61].

Auto ARIMA:

In time series the correct way of processing data is to fit different models to data and make a choice. Auto-ARIMA offers a way around this through the use of an automated process which tests through all possible orders and parameters of the model until it has adopted the best setting based on statistical indicators. It identifies the optimal likelihood measure that addresses data loss and simplifies the AIC and BIC indicators for model complexity control. However, when dealing with powerful models it is important to know that the process of iterating through models can take time. Auto-ARIMA is much better in that it automates the model selection process to give the best fitting model for forecasting and analysis. The formula for Log-Likelihood, AIC and BIC is given below in equations 1, 2, and 3.

$$-2 \ln \ln \lambda = 2 \sum_i O_i \ln \ln \left(\frac{O_i}{E_i} \right) \quad (1)$$

$$AIC(k) = -2 \log \log (L) + k \times edf \quad (2)$$

$$BIC = -2 \log \log Likelihood + \log \log (n) p \quad (3)$$

Pros and Cons of Auto-ARIMA:

Pros:

The advantages of this automated model are to the use of the automated model selection technique.

- It particularly minimizes the amount of time which is taken in choosing a model.

- It frees the data scientist from having to know the statistical theory of each model and choosing from them.
- It also reduces the likelihood of human mistake in perusing as well as implementing various selection parameters.

Cons:

Although there are certain advantages of full automating some risks are also there.

- Using the automated criteria only means that some potentially useful parameters for (p, d, q) will not be included in the analysis.
- It does not facilitate how data scientists can look at different models with some sort of ranking to perform a comparison.
- But it is an inadequate replacement for the knowledge of the data and the field that model selection requires.

Methodology for Global Renewable Energy Production (GREP) (2000-2023)

Dataset:

The main purpose of this research is to conduct the correlation analysis, or find the similar trends between similar columns of Wales and global energy dataset. In this way, a dataset downloaded from Kaggle [62]. This dataset captures patterns of renewable energy production considering timescales as well as countries. The set of information contains yearly data from 2000 to 2023. Renewable sources here include solar, wind, hydroelectric and other renewables like geothermal and biomass. This data is presented country by country and year by year. The data presented include production in gigawatt-hours with regards to solar, wind, hydroelectric, other renewables, and total renewable energy for every country and calendar year.

The objective of the dataset is to facilitate understanding of trends in renewable energy generation and produced capacities by type and region. In so doing it can give an understanding of the trends of renewable energy as adopted in the countries and across the world for the past two decades. It can be used to compare one country's source mix to another's to discover nation's leading in production capacity of certain technologies such as photovoltaic solar and wind power turbines. Taken together, these data can be leveraged in quantitative research and modeling for the continuing transition to renewable energy on a global scale.

In the dataset preprocessing step, it passes through all pre-processing steps like Wales data including checking for missing and duplicate values in the dataset to ensure cleanliness for forecasting models given in Figure 19.

```
Missing Values:
Year          0
Country       0
SolarEnergy   0
WindEnergy    0
HydroEnergy   0
OtherRenewableEnergy  0
TotalRenewableEnergy  0
dtype: int64

Duplicate Rows:
0
```

Figure 19: Missing and Duplicate values in GREP Data

After that, the distribution of columns in the dataset are plotted to see their overall column distribution and detect outliers using box-plot to enhance dataset quality. Furthermore, box plots are also created to illustrate the distribution of different types of renewable energy including solar energy, wind energy, hydro energy and others by country mention in Figure 20. To determine the fluctuations and anomalies of energy production so that they can be compared across the various countries. By applying such a strategy, this will present a factual picture of how the various nations are employing renewable energy sources and possible similarities or differences observable on the energy supply trends.

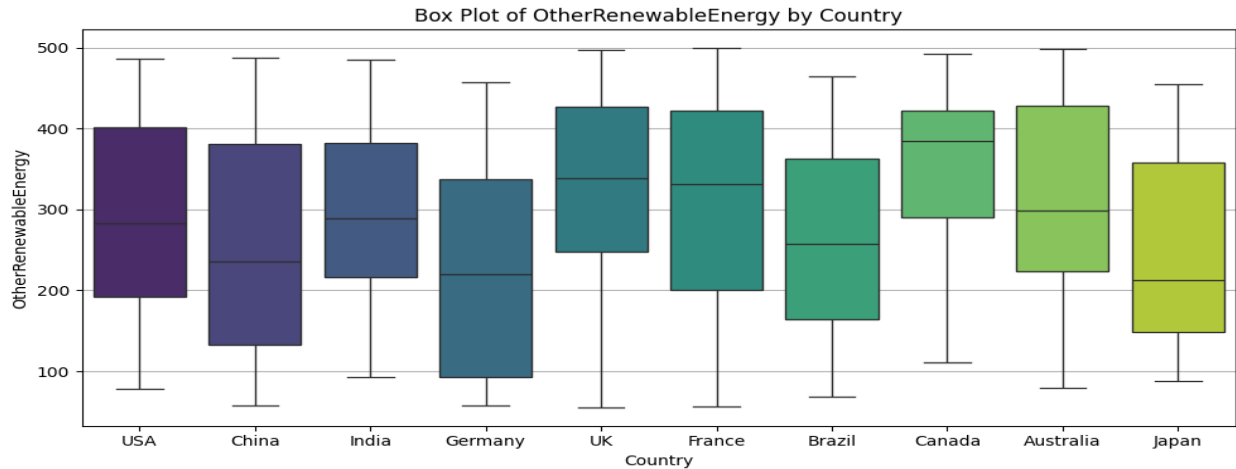


Figure 20: Box plot for Other Renewable Energy by Country

Finally, like the Wales based time series forecasting to estimate potential future generation of renewable energy among others; solar energy, wind energy, hydro power and other renewable forms of energy are forecasted for next 3 years. The data has been summed up by year to study the general energy tendencies and the auto_arima model is used to predict the energy production for the following years. Again, the predicted and past values are used to graphically present the future trends or the changes in the use of renewable energy to show areas that may have growth or areas where difficulties might be experienced in achieving sustainable energy objectives.

Correlation Analysis between Wales Data & GREP Data:

In the final part of methodology, both Wales & GREP datasets merged to combine renewable energy information using the year to enable a correlation analysis of different energy parameters. Numerical columns for correlation analysis are introduced: electrical capacity, heat capacity, electrical generation, and various types of renewable energy sources. Pearson correlation coefficient is then computed to analyze the patterns that exist among these numerical predictors and quantify how various types of renewable electricity generation and capacity are connected to one another. By calculating the correlation matrix, this research can detect high correlations that can be useful to promote strategic objectives in energy production and policy formulation.

4. Results and Discussion:

The main purpose of this project was to use statistical trend analysis, and data-driven approaches to forecast the electrical capacity, and electrical generation in regions using time series models. In this way, initially dataset for the Wales data collected from the Stats Wales under catalog for environment and countryside energies section. It contains different columns including Number of Projects, Electrical Capacity, Heat Capacity, Electrical Generation and Heat Generation. During the EDA part of data cleaning this found that projects, electrical and Heat capacity, Electrical and Heat Generation contain a list of 84, missing values in the same rows as visually represented in Figure 8. So, after checking the distribution of columns missing values filled with 0 because, mode was 0 and also mean was in this range. Going further insight into data distribution of columns was checked and found the dataset is highly right skewed and contains a list of outliers. That can cause the problem of giving accurate forecasting results. To avoid any issue Yeo-Johnson transformation was used to retransform the Wales dataset from a right-skewed distribution to a distribution closer to the normal distribution in order to improve the results of further statistical testing and machine learning algorithms. Yeo and Johnson transformation is best because it provides both positive and negative values as opposed to others like logarithmic and Box-Cox. From this transformation outliers in the dataset removed as given in Figure 12 and 13.

4.1. Observation for Technologies vs. Electrical Capacity:

After that, box plots and KDE distribution plots give a view of the spread and distribution of electrical capacities for various RE technologies. For instance figure 21, shows that the maximum electrical capacity attainable is nearly 1.5 MWe, and through the distribution plot, one might note that it is right-skewed. There is a relatively large number of installations with low electrical power ratings of 0–0.5 MWe with a few installations with much higher power ratings.

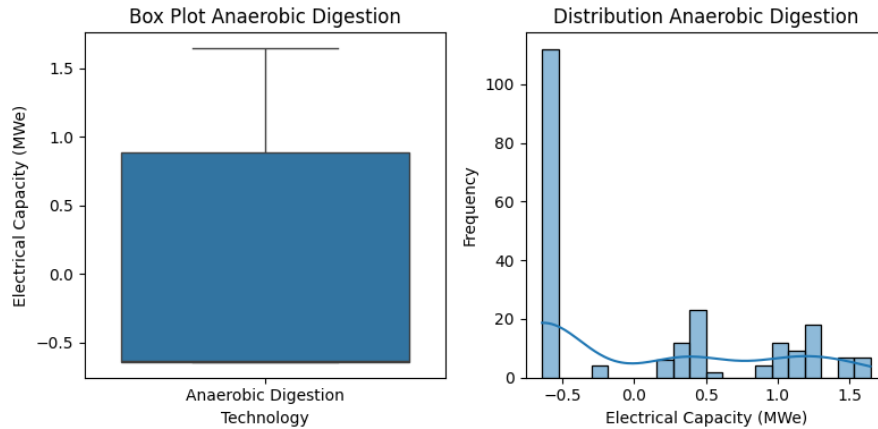


Figure 21: Plot for Anaerobic Digestion

Similarly, Energy from Waste Technology has few plants with slightly higher electrical capacities while many operate at near zero or slightly negative. The frequency distribution is Positively skewed, and such a distribution of installation capacities clearly shows that there are many small clients, and only occasionally clients with large installation capacities.

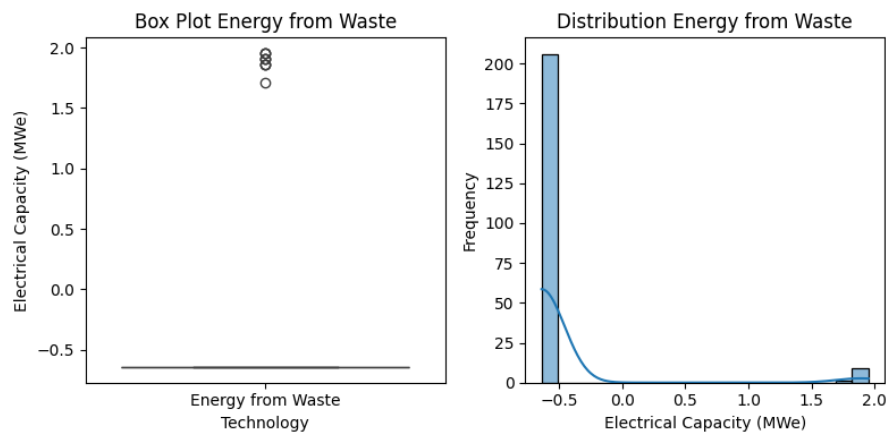


Figure 22: Plot for Energy from Waste

Considering the heat pump technology there is very little spread in the electrical capacity of the results as evidenced by the box plot that has a very narrow whisker bar at about -0.6MWe. The distribution plot reflects this, and is symmetrical with a clear peak at about -0.6 MWe to reflect the fact that heat pump installations have a relatively uniform capacity.

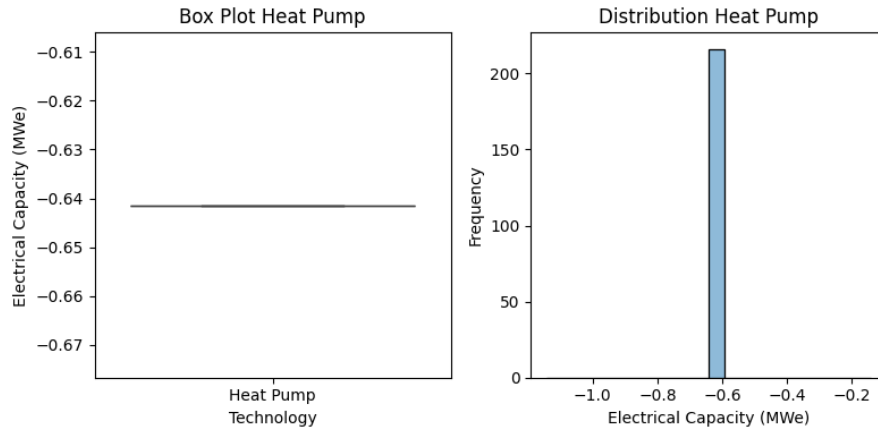


Figure 23: Electrical Capacity vs. Heat Pump

The values comparing hydropower have bigger IQRs than those of wind and solar and very close to the median values that are located at 0 MWe. The distribution is extended over a range of negative to positive, which seems to indicate the variation of installations has a good diversity. While the majority of, 75% of the hydropower capacities range from -0.5 to 0.5MWe, the plot also reveals outliers going up to 2 MWe.

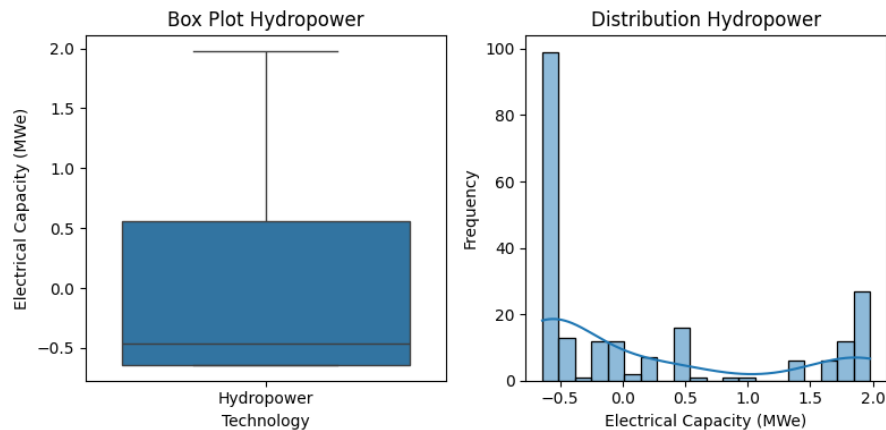


Figure 24: Electrical Capacity vs. Hydropower

In this offshore wind electrical capacities are close to -0.5 MWe as depicted in the box and distribution plot. There is one exception here with the electrical capacity close to 2 MWe which is conditioned by the fact that there are more capacious installations of offshore wind power.

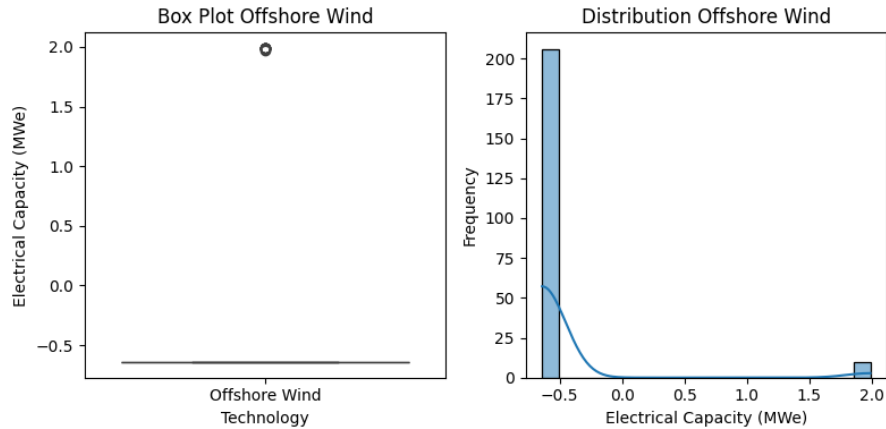


Figure 25: Offshore wind vs. Electrical Capacity

4.1.1. Summary Electrical Capacity vs. major Technology:

It is also rather apparent that each technology has its individual fluctuation of capacities within the electrical power sector, with Anaerobic Digestion and Biomass Electricity and CHP with rather narrow average, but significant variations from average values; Heat Pump and Nuclear demonstrates higher homogeneity with fewer extreme values in the middle of the range. They give information on the current size and layout of various renewable energy technologies on a regional level, thereby providing an assessment on the prospects for further extension and efficiency based on the observed capacity margins.

4.2. Observations for Technologies and Regions:

After getting insights into, electrical capacity vs all technologies in these steps, technologies and region comparison were made, indicates that, for instance, the boxplots of the electrical capacity (MWe) of different energy technology in Wales regions, indicate regional differences in electrical capacity, whereby capacity of certain technologies ranges widely (e.g. Onshore Wind) while that of others is either negligible or absent in some regions. Outliers are indicated by small circles beyond the whiskers mostly for unknown technology.

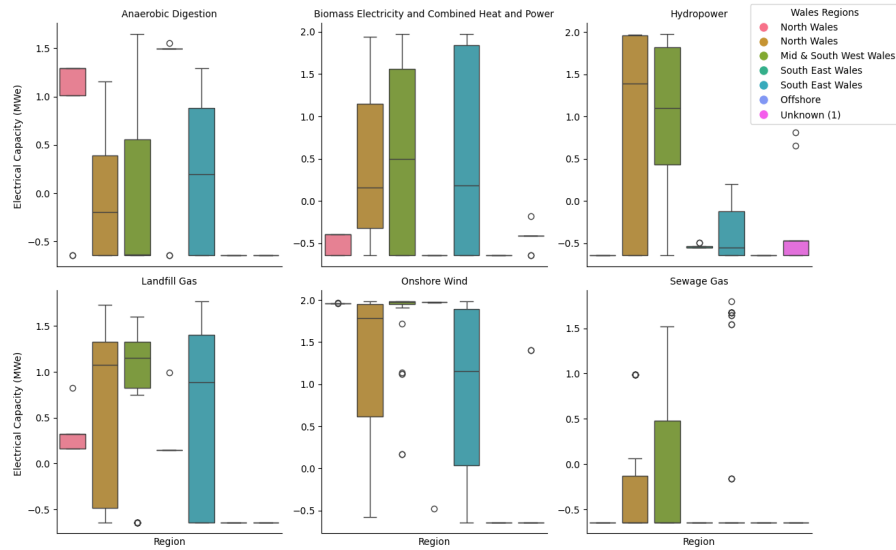


Figure 26: Box-plot for Electrical Capacity for different regions:

For Heat Capacity on different technologies and different regions in Wales, box plots indicate from Figure 27, for Hydropower, there are no variations at all in data points, as presented by a flat straight line across the regions. It is clear that Mid & South West Wales has a higher capacity for some forms of Anaerobic Digestion and Biomass Electricity, as is indicated by the boxplots. In Biomass Electricity and Sewage Gas categories are considered as outliers. It is generally observed that, in relation to different energy technologies in Wales, general heat capacities have diversified between increases of large magnitudes and those which remained small or constant, or even minimal.

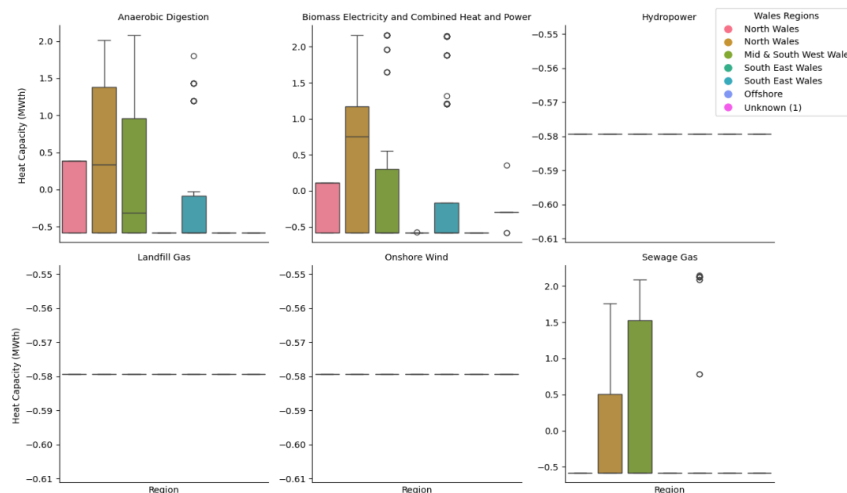


Figure 27: Box-plot for Heat Capacity for different regions

Last but not least, for electrical generation in the Mid & South West Wales and South East Wales, there is higher and more variable generation in Anaerobic Digestion and Biomass Electricity. Hydropower remains almost constant while Landfill Gas fluctuates highly, most notably in South East Wales. Onshore Wind has several anomalies mainly in North Wales and in South East Wales which depicts regional trends. There is higher Sewage Gas generation in Mid & South West Wales as well as few of the bar is showing higher values.

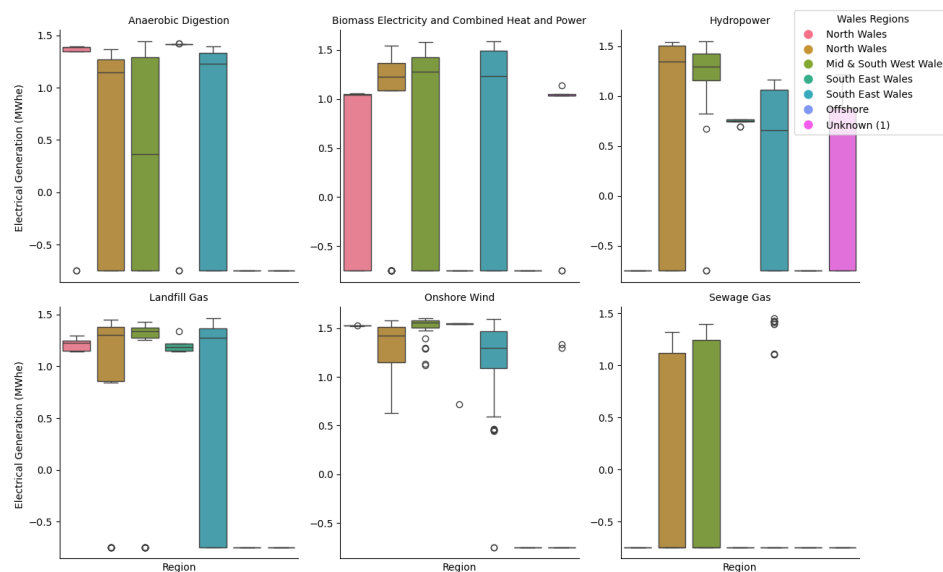


Figure 28: Box-plot for Electrical Generation for different regions

Finally, the average heat generation by regions with slight variation in Anaerobic Digestion, Biomass Electricity, and Sewage Gas. Some of the sources such as Onshore Wind, Landfill Gas and Hydropower have displayed either no or insignificant heat output. We can observe that Sewage Gas and Biomass Electricity sources show variability across some regions represented by the error bars, while Hydropower has regions which are closer to each other on the map.

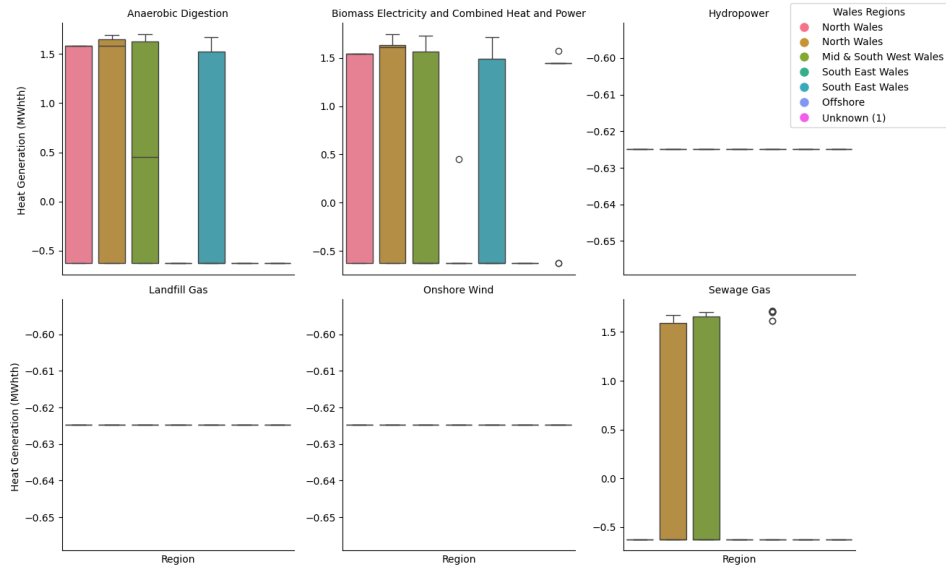


Figure 29: Box-plot for Heat Generation for different regions

4.2.1. Summary Technologies vs. Regions:

This study of renewable energy technologies by region shows that there are large variations between regions in electrical capacity, heat capacity, and electricity generated. Boxplots reveal that although there are wide variations in electrical capacity of some technologies like Onshore Wind, there are very little data points in technologies like Hydropower implying their limited deployment in some regions. Specifically, Anaerobic Digestion and Biomass Electricity show greater and fluctuating levels of generation in Mid & South West Wales and South East Wales. Hydropower is steady in comparison while Landfill Gas varies considerably, especially in South East Wales. The average heat generation means data show the fluctuations; some technologies are even appreciable in terms of heat generated which are otherwise very low like Onshore Wind and Hydropower. In general, the results highlight the disparities in energy generation and potential in the regional setting of Wales with regards to different types of renewable technologies.

4.3. Results for Correlation and Trend Analysis among Wales dataset columns:

After deep insights into technologies, Regions of dataset, in this step when correlation analysis is performed to see the impact of one parameter to another parameters. For instance, figure 30, highlights that, Heat capacity positively correlates with heat generation and similarly electrical capacity with electrical generation both displaying positive coefficients of 0.92 and 0.91 respectively

which shows that where capacity is large for heat and or electricity, the generation will also be large. It also shows that the number of projects has a moderate positive relationship with heat capacity $r = 0.55$ and heat generation 0.52 but a much lower positive relationship with electrical capacity and electrical generation. There is a negative relationship between electrical and heat characteristics which imply that there is some compromise between electrical capability and heat capability and generation. Altogether, Heat and electrical capacities bear strongest relation to their generation and amongst both the heat capacity and generation are highly correlated with coefficient of 0.92 . Electrical and heat variables are shown to have weak negative relationships with one another, meaning that increase in one decreases the other.

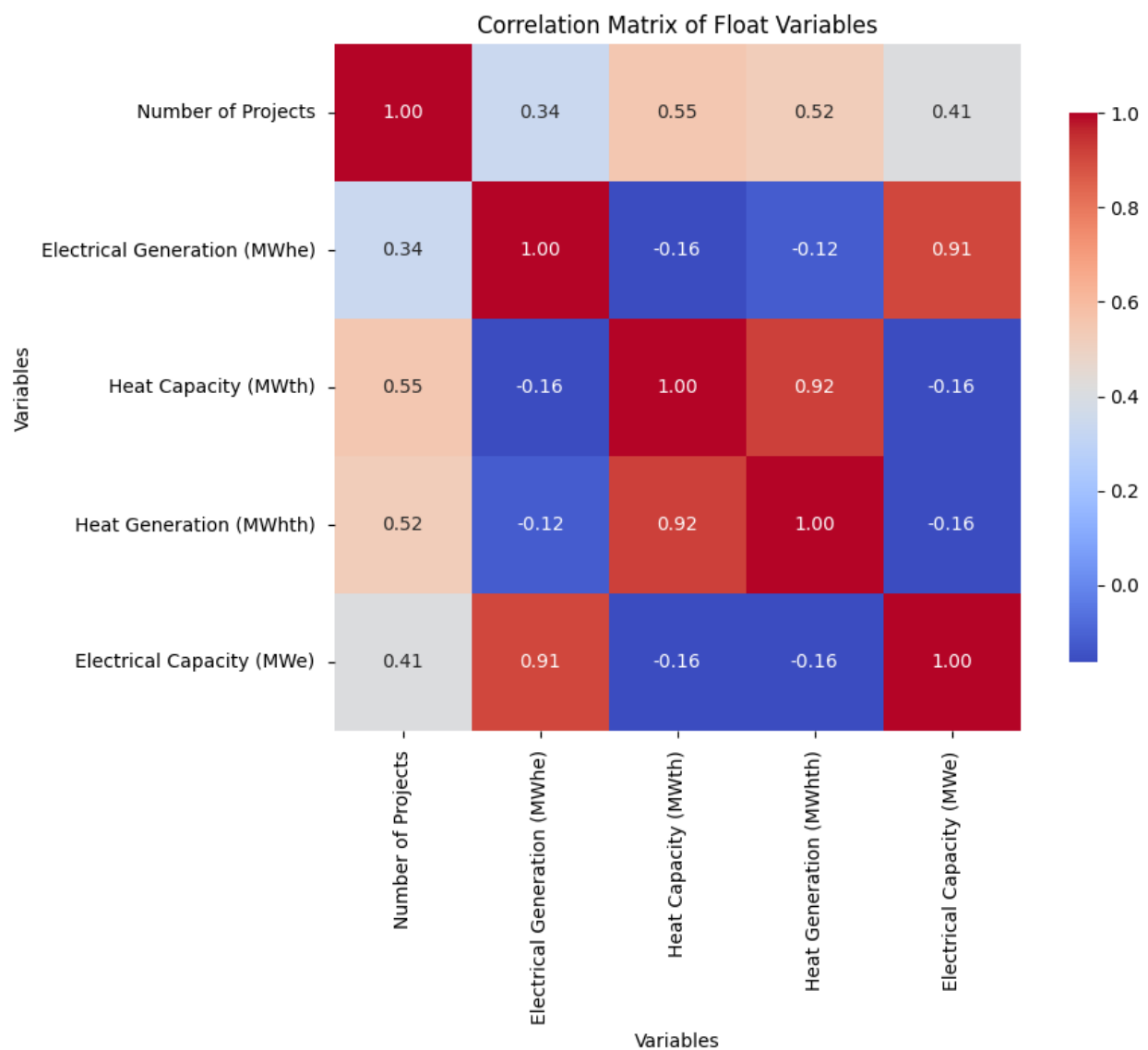


Figure 30: Correlation analysis among Wales data columns

In order to further prove the correlation analysis, trend analysis performed between Electrical Capacity and Electrical generation in Figure 31. The trend analysis reveals that over the last decade from 2012 to 2022, electrical generation and electrical capacity rose continuously from year to year. In the beginning, installed capacity expanded more rapidly than generation, although in the year 2016, generation capacity seemed to match the expansion in installed capacity. They were however relatively stable and rose from 2016 to 2022 in parallel to each other with slight differences. This implies that the more capacity that was developed, the more the energy produced in proportion to the increase in the capacity. In other words, there is a clear correlation between electrical generation and amount of capacity; and both have been on the rise in recent decades.

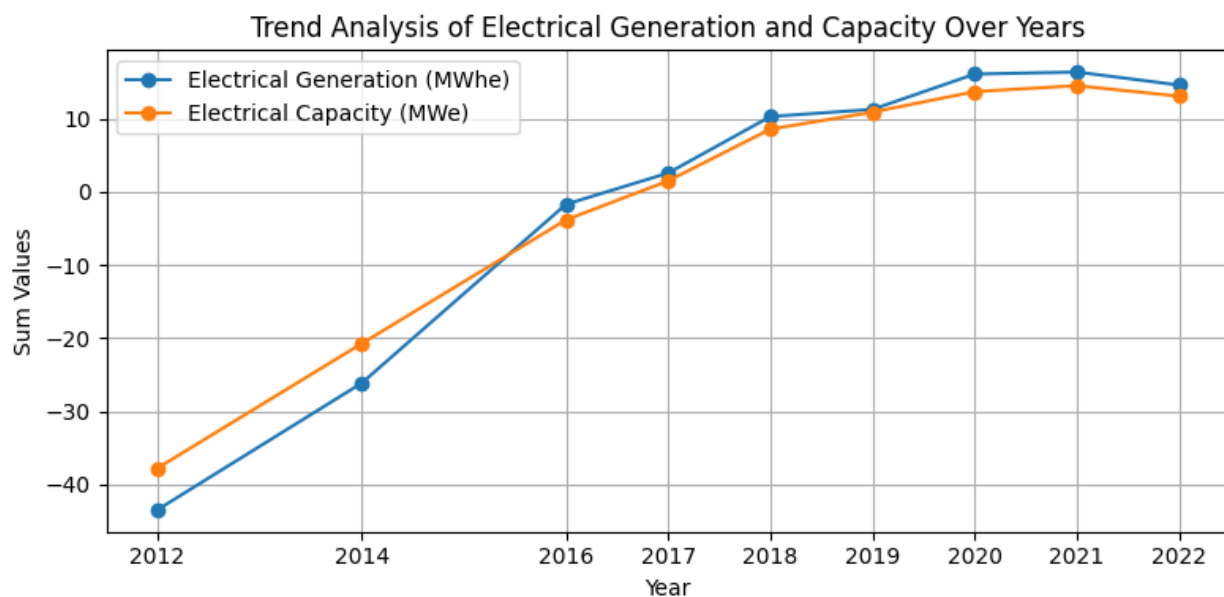


Figure 31: Trend Analysis Electrical Generation vs. Electrical Capacity

4.3.1. Rational For Negative Values

The negative values noted in the graph could be attributed to the change made to the data set, the kind of transformation that was made that generated the results on the human development index. In particular, Yeo-Johnson transformation was applied to solve the missing values as using means to impute these records ended up in outliers. It is likely that outliers were the major contributing factor that skewed the data distribution hence when the data was transformed using the Yeo-Johnson transformation, these values were reduced to negative ones. In this way, the transformation made it possible to maintain the stability of the ratios between different variables for electrical generation and capacity over the years, and rid the data of the negative consequences of missing values and higher outliers.

4.4. Forecasting Results and Selection of Auto-ARIMA over other Models:

While comparing annual data, Auto-ARIMA gives a simple edge in estimating the most appropriate ARIMA parameters for a given series in addition to handling both seasonal and non-seasonal data. Some of the features used in this model include differencing and model selection criteria which are better than manually searching for the right parameters. As a result, Auto-ARIMA dramatically brings down the likelihood of a human mistake and time spent on model selection while guaranteeing that selected model adheres to the underlying data patterns and trends properly. Auto-ARIMA is therefore more efficient than other methods, and has the advantage of being highly reactive depending on time series characteristics; thus, is best suited for the yearly time series data.

Auto-ARIMA, however, has some benefits when compared to Prophet for time series forecasting. First, it fits the statistical properties of the data well due to its autoregressive and moving average components, which are ideal on a dataset with marked temporal structure. Also, Auto-ARIMA provides the best choice of parameters, which means that the user is able to avoid frequent adjustment and obtain the ideal model fitting easily. Although Prophet Recommendation adapts easily to seasonality and holidays, Auto-ARIMA gives a more precise result for data with autocorrelation and stationary nature, especially where patterns are not complex. Furthermore, the statistical approach utilized in Auto-ARIMA provides better results when dealing with comparatively small datasets, which can be a drawback for Prophet, that often needs more data to take advantage of its functionalities.

So, after the advantages of Auto-ARIMA over other models when employed in this research, I got the following results, of Best Model parameters as given in Figure 29. This indicates that, stepwise approach of the search for the minimum AIC successfully implemented in the identification of the appropriate ARIMA model for the dataset. Basic models such as ARIMA(0,0,0)(0,0,0)[0], produced higher AIC measures of around 74 where as expanded models such as ARIMA(1,0,1)(0,0,0)[0] yielded much lower AIC of 74.038 at the outset. During the first round the model that turned out to be the best with an AIC of 71.945 was the ARIMA(2,0,0)(0,0,0)[0]. In another search of the current data, this model was confirmed to be the most appropriate given its lower AIC of 64.789.

```

Performing stepwise search to minimize aic
ARIMA(1,0,1)(0,0,0)[0] : AIC=74.038, Time=0.30 sec
ARIMA(0,0,0)(0,0,0)[0] : AIC=81.357, Time=0.03 sec
ARIMA(1,0,0)(0,0,0)[0] : AIC=74.669, Time=0.04 sec
ARIMA(0,0,1)(0,0,0)[0] : AIC=inf, Time=0.07 sec
ARIMA(2,0,1)(0,0,0)[0] : AIC=73.628, Time=0.27 sec
ARIMA(2,0,0)(0,0,0)[0] : AIC=71.945, Time=0.07 sec
ARIMA(3,0,0)(0,0,0)[0] : AIC=73.512, Time=0.11 sec
ARIMA(3,0,1)(0,0,0)[0] : AIC=75.036, Time=1.25 sec
ARIMA(2,0,0)(0,0,0)[0] intercept : AIC=73.477, Time=0.58 sec

Best model: ARIMA(2,0,0)(0,0,0)[0]
Total fit time: 2.747 seconds
Performing stepwise search to minimize aic
ARIMA(1,0,1)(0,0,0)[0] : AIC=68.494, Time=0.04 sec
ARIMA(0,0,0)(0,0,0)[0] : AIC=78.662, Time=0.01 sec
ARIMA(1,0,0)(0,0,0)[0] : AIC=71.238, Time=0.02 sec
ARIMA(0,0,1)(0,0,0)[0] : AIC=inf, Time=0.04 sec
ARIMA(2,0,1)(0,0,0)[0] : AIC=66.789, Time=0.06 sec
ARIMA(2,0,0)(0,0,0)[0] : AIC=64.789, Time=0.03 sec
ARIMA(3,0,0)(0,0,0)[0] : AIC=66.789, Time=0.05 sec
ARIMA(3,0,1)(0,0,0)[0] : AIC=68.726, Time=0.09 sec
ARIMA(2,0,0)(0,0,0)[0] intercept : AIC=65.329, Time=0.07 sec

Best model: ARIMA(2,0,0)(0,0,0)[0]
Total fit time: 0.423 seconds

```

Figure 32: Parameter Selection for Auto-ARIMA

The trends of both generation and capacity are almost the same during the years. Both values were initially negative from 2012 and declined to the lowest level of approximately – 40 MW in 2012. From there, both metrics increased and rose to slightly above 10MW in 2019, the data also show. Both curves declined after 2019 and trended from 2020 slightly though the time series was not shown in the figure 33. This forecast also begins in the year 2022 marked by the vertical dashed line here as evidence that both the electrical generation and the capacity are expected to reduce slightly year by year in the forecasted period of 2022 to 2027. This downward trend indicates that the infrastructure of the electrical industry may decrease its efficiency in the future years or there may be a decrease in demand/supply in the years ahead of the forecast. The parallelism emphasized that electrical generation strongly relates to electrical capacity in the past and in the future numerical figures.

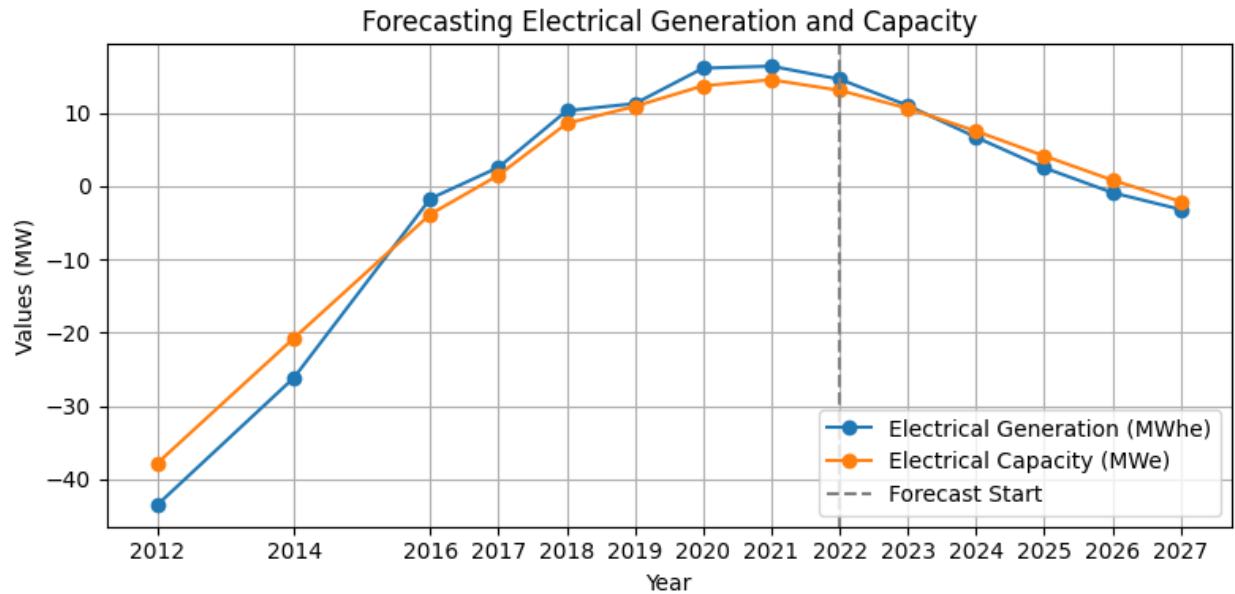


Figure 33: Forecasting Electrical Generation vs. Capacity (2022 – 2027)

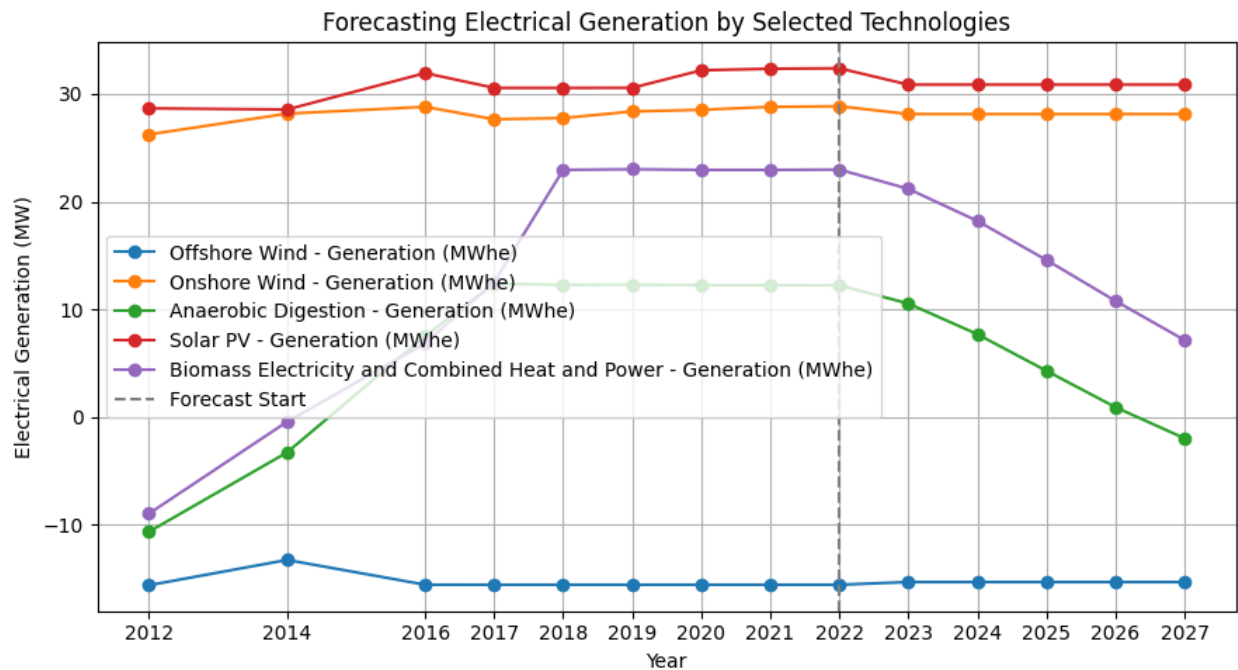


Figure 34: Electrical Generation by Technology

The graph shown in Figure 34, the electrical generation forecast of the technologies for the years 2022-2027, and the historical generation statistics in the years 2012-2021. Solar PV generation always has the highest generation values, the highest being approximately 30 MW which remains almost consistent throughout the historical period and the forecast. Onshore wind similarly exhibits a steady succession maintaining just below solar PV generation at about 25MW. Biomass

electricity and CHP have a growing scale from 2015 to reach 17-20MW in 2021 but will decline over the analysis period. Green line represents anaerobic digestion, which also grows continuously to 2021, approximately 10 MW and then declines during the forecasting period.

The trends further depict that solar PV and onshore wind will remain the leading technologies for electrical generation of among the above-mentioned selected technologies, with relatively steady-generation throughout the forecast period. However, the reduction of biomass and anaerobic digestion might be an issue under consideration, since it tends to reduce diversity in renewable energy. The periodic movement in the generation of solar PV and onshore wind power clearly show their stability and their ability to provide good returns that's why their Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plotted given in Figure 35 and 36.

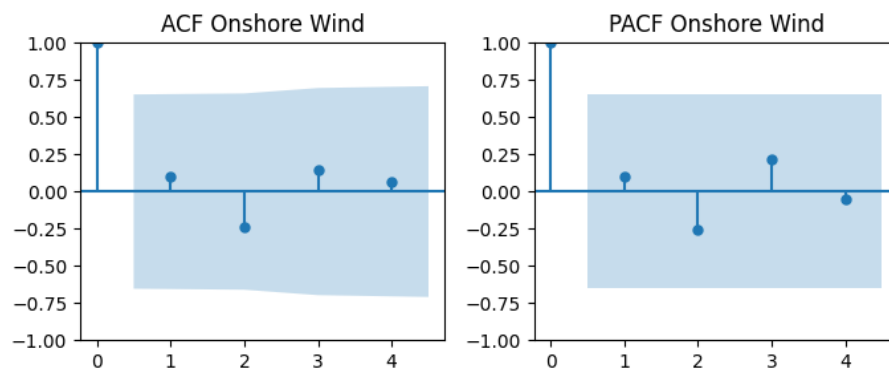


Figure 35: ACF- PACF for Onshore Wind

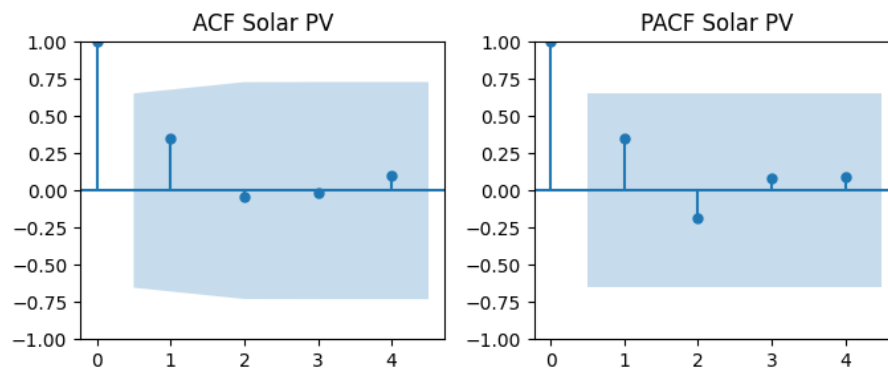


Figure 36: ACF- PACF plot for Solar PV

In order to further get insights into time series models parameter, ACF and PACF graph to identify the order of the ARIMA model by diagnosing the right parameters for sales time series. These plots assist in deciding the order of the integrated AR and MA models after observing the strength of correlation in the identified lags. From Figures 32, 33 this can be seen that the results obtained

from the ACF and PACF of Onshore Wind where the graphs exhibit a similar trend to that of the Solar PV charts but with slight differences. ACF plots are significant for the first lag, which means strong correlation between the current period and previous period data and then rapidly decreases, which also indicates the weak correlation between the consecutive lags. Just like in the case of Solar PV, the first lag reigns supreme in the PACF, but lags 3 have a stronger correlation, which is due to the persistence of some residual response to earlier values. In this case, the PACF plot of Onshore Wind is less stable and more unsmooth compared with that of Solar PV which may indicate a more complicated time series characteristic of Onshore Wind.

4.5. Results for GREP:

Similarly, like Wales dataset, here for GREP dataset collected, preprocessed to make it ready for comparison with Wales data. However, this noted that the dataset does not have any missing and duplicate values and almost the whole dataset distribution is like the uniform distribution that can be seen given in Figure 37.

The key observations highlighted here show variations of global renewable energy by sources in the period between 2000 and 2023. The solar energy production ranges between 350 and 800 with the median of 600 units, while the wind energy potential is ranging in-between 400 and 1200 with the median of 1100 units. As has been mentioned, hydro energy is fairly stable, normally oscillating between 1000 and 1750 units. Other renewable sources also have the least production capacity with a median of 350 and IQR of 250-400 capacity units. The graph of total renewable energy production is the most consistent of all the graphs with a median of 3000 units and an ideal bell-shaped curve illustrating that values are oriented more towards the center of this number. Alternatively, the share of wind, hydro, and solar energy account for the majority of the renewable production where there is high volatility in hydro and wind energy production; nevertheless, total renewable energy has relatively constant rate in production.

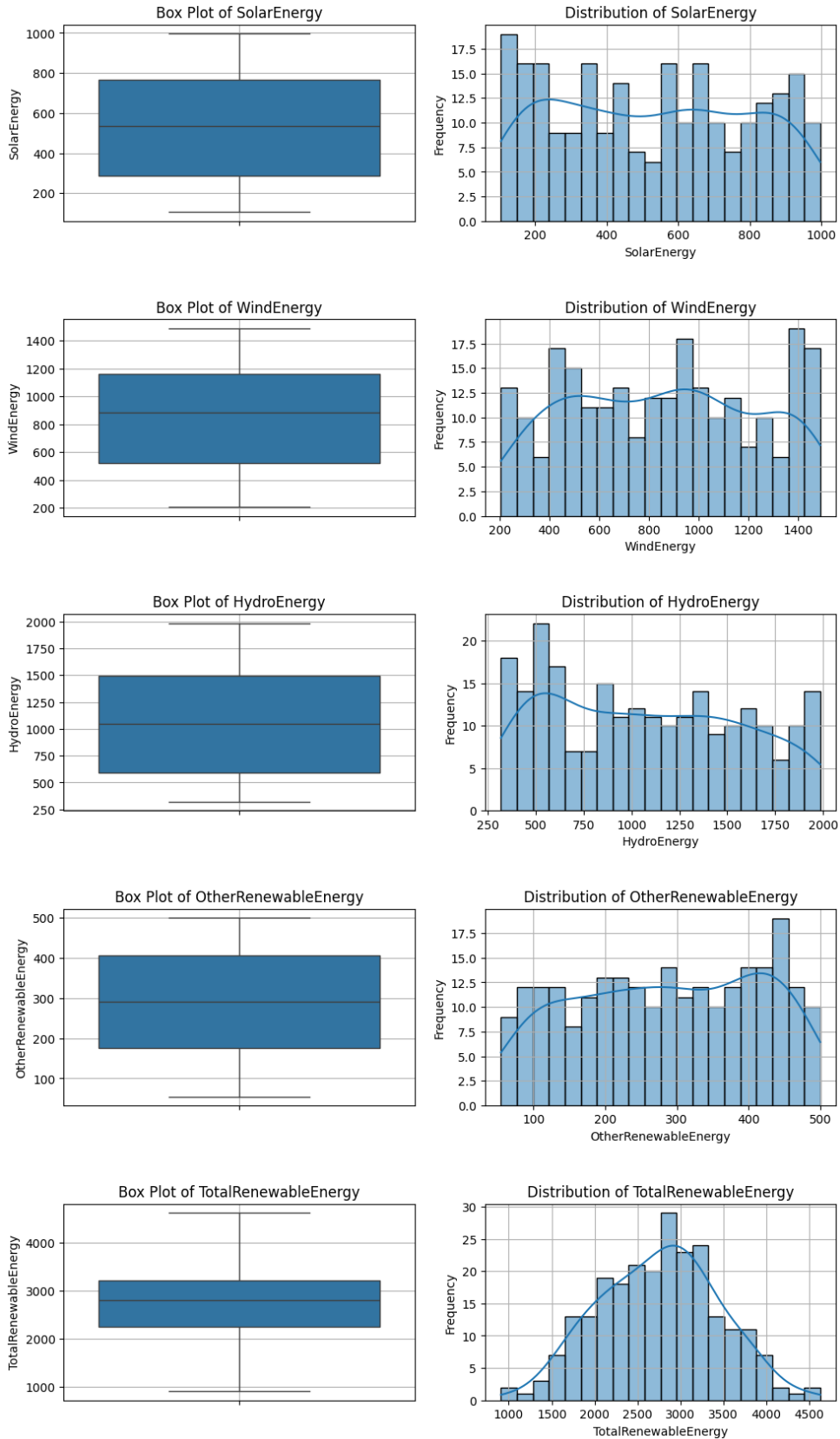


Figure 37: Distribution of Variables in GREP

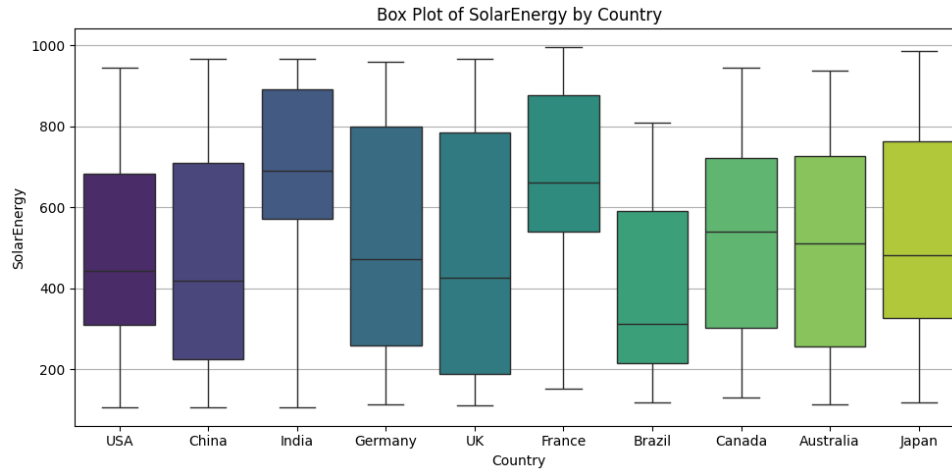


Figure 38: Solar Energy by Country

To understand which country has the most even distribution of solar energy this means that out of all median values of energy distribution for countries in the world France has the highest. The dispersion of this solar energy generation is much higher in India and France than in other countries since it is much spread-out of the scale. On the other hand, Brazil shows a lower solar energy capacity that fluctuates very little while presenting less capacity, but steady percentages.

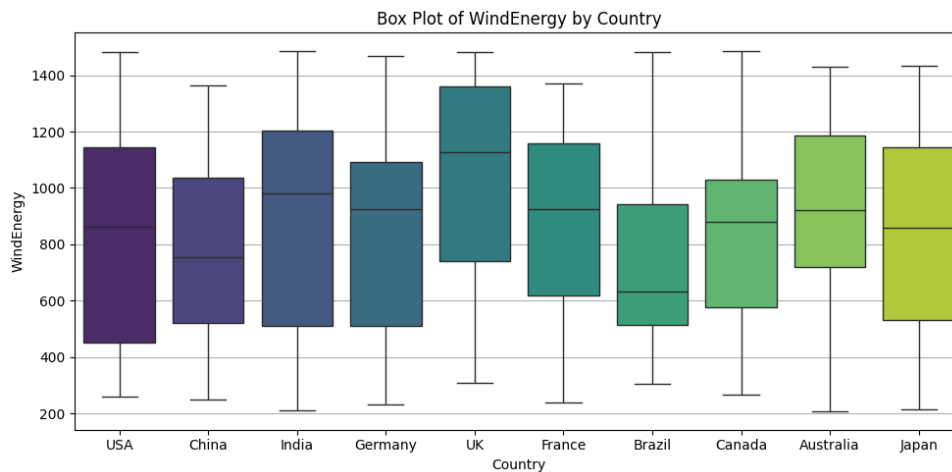


Figure 39: Wind Energy by Country

For Wind energy by country. UK has the highest median capability, still, some instability exists within the range of generated wind power. Australia and Japan have moderate wind energy potentials but their volatility proves that conditions in each region differ.

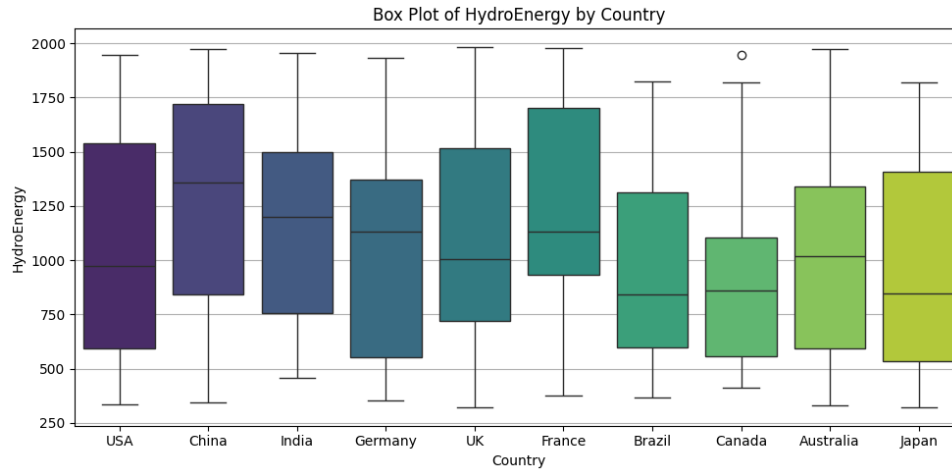


Figure 40: Hydro Energy by Country

Hydropower generation again seems to be more consistent by country, which can be seen by their median, USA, China and France are leading. Brazil has recorded low levels of hydro energy generation with a small interquartile spread showing fairly consistent but low levels of generation. Canada has one outlier in the set indicating one single case of very high hydroelectric output in comparison to other values typical of Canada.

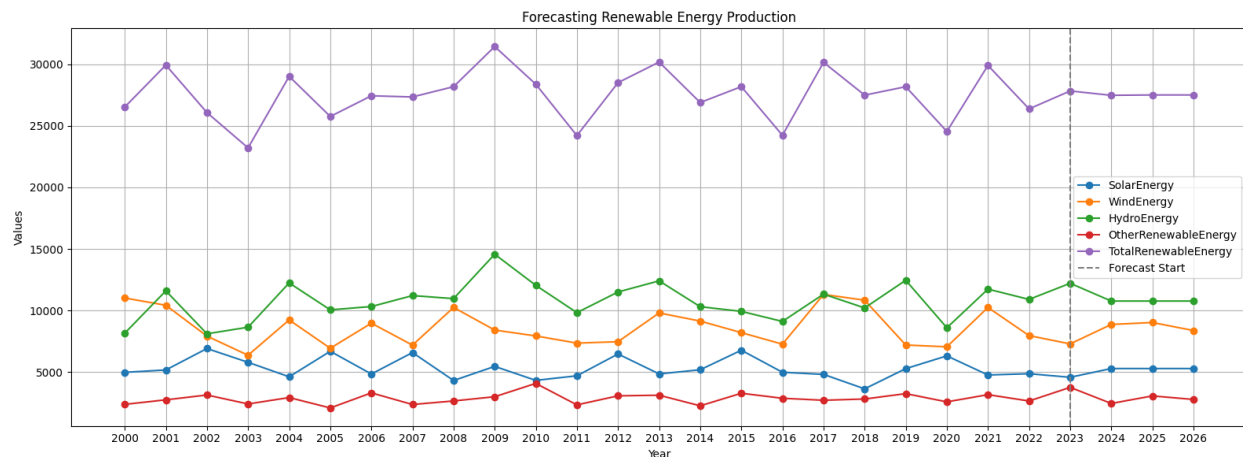


Figure 41: Forecasting Renewable Energy Production UK (2024 – 2026)

From the provided graph forecasting renewable energy production, key findings and observations for the next three years (2024–2026) are as follows:

1. Total Renewable Energy: Their combined totals show that global renewable energy production is weak and in limited fluctuation for the upcoming three years. The forecast shows that the total generation of renewable energy will remain almost at the level of the past few years at 27000 to 30000 units.

2. Solar Energy: Looking at the use of solar energy, there is a gradual, marginal rise in the numbers year by year and the projections are that it will remain the same. Production in the generation of solar energy is expected to increase slightly by 2026.
3. Wind Energy: The pattern of generating wind energy seems relatively flat with small fluctuation in the production during the years. According to the wind energy forecast, the level of wind energy production is expected not to fluctuate significantly in the future to remain at this level in the next three years.
4. Hydro Energy: Hydro energy actually shows some unstable or fluctuating trend in the past years but in recent years it has manifested a stable trend. As to the forecast for the volume of sales, the influx of hydro energy production should flatten in the range of 8,000 – 11,000 units.
5. Other Renewable Energy: Finally, this energy category currently produces very low levels and there has been little fluctuation in the production over the years. The forecast is also in the same direction that there would not be a great deal of change in the near future.

In sum, the total renewable energy is expected to remain static with a slight increase in the solar energy and other renewable energy power such as wind and other renewable energy are expected to follow the current trend with minor oscillation over the next three years.

4.6. Results for Correlation Analysis between Wales and GREP:

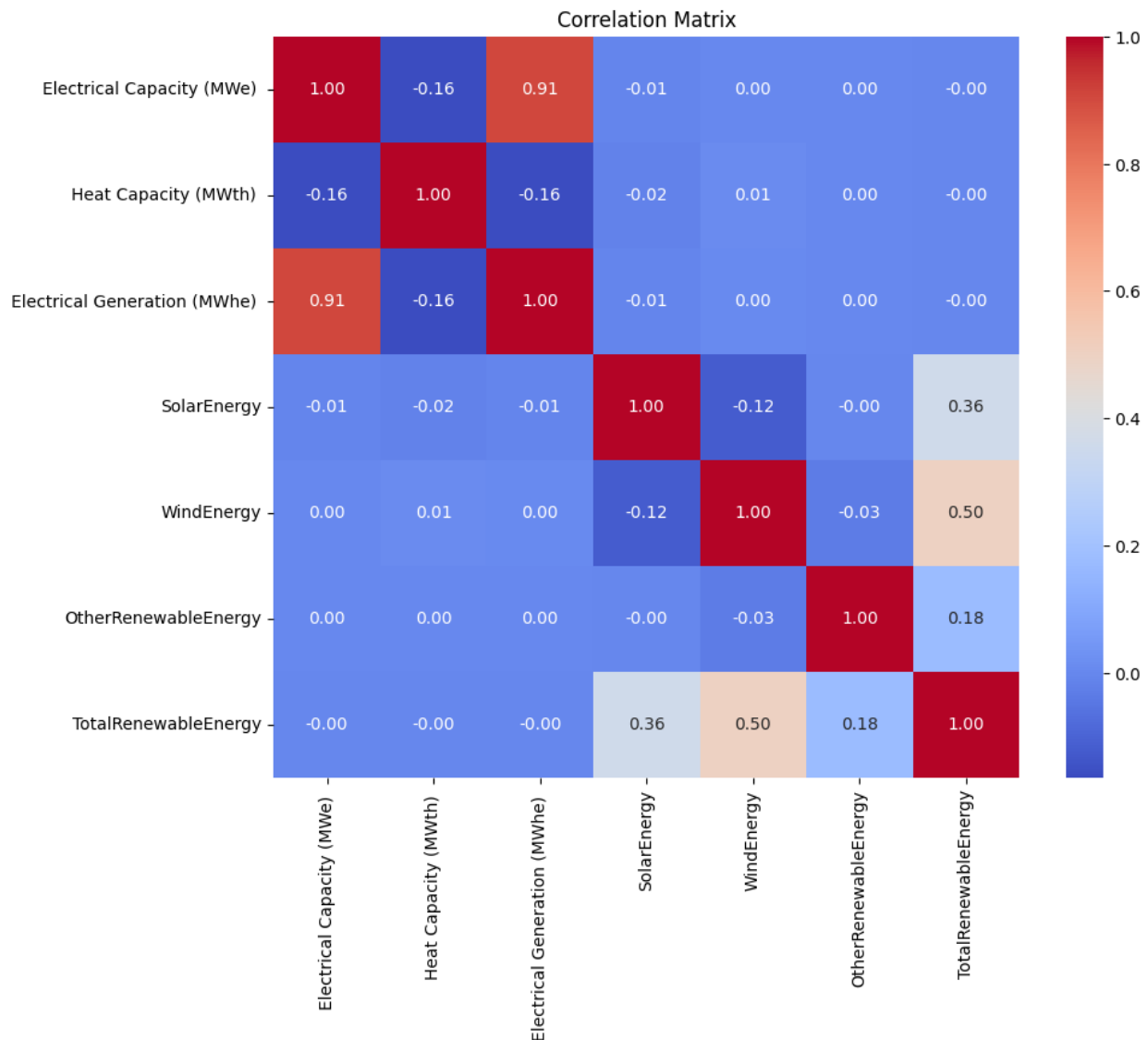


Figure 42: Correlation Matrix for Renewable Energy Wales vs. GREP

The correlation matrix combines two datasets: One for Wales which specializes in electrical and heat capacities; the second one is Global Renewable Energy Production which specializes in different types of renew-able energy. This matrix from Figure 42, shows that there is a very weak positive relationship or none at all between the two datasets. The coefficients of determination from the Wales dataset like Electrical Capacity, Heat Capacity, and Electrical Generation are very close to zero with world renewable energy production indicators like Solar Energy, Wind Energy, Other Renewable Energy, and Total Renewable Energy.

For example, the Electrical Capacity is still moving almost independently of any global renewable energy variable ranging between -0.01 and 0.00, meaning that any changes in Wales capacity or generation are not closely connected with the global trend in renewable energy.

Likewise, Heat Capacity and Electrical Generation have little correlation with renewal energy output globally. None of the correlations apply to both datasets; the ones calculated are limited by each dataset: Electrical Capacity and Electrical Generation in Wales are 0.91; Wind Energy and Total Renewable Energy – global are +0.50. Although there is a generally positive relationship of energy capacities and generation in Wales, there is no significant variability correlation between variability of the energy capacities and generation in Wales, and the variability of the global renewable energy production from the year 2000 to 2023.

4.7. Dashboard Creation

In the final part of methodology, dashboard results for Wales region and GREP are given in Figure 43 and 44 respectively. For instance, the Wales region dashboard gives basic data on Wales' regional electrical energy initiatives, capabilities, and production analytical data. The data is divided by sub-regions and counties of Wales, which means that the user can choose what information he needs, or what year is needed.

That provides the visual description of different energy features. For instance,

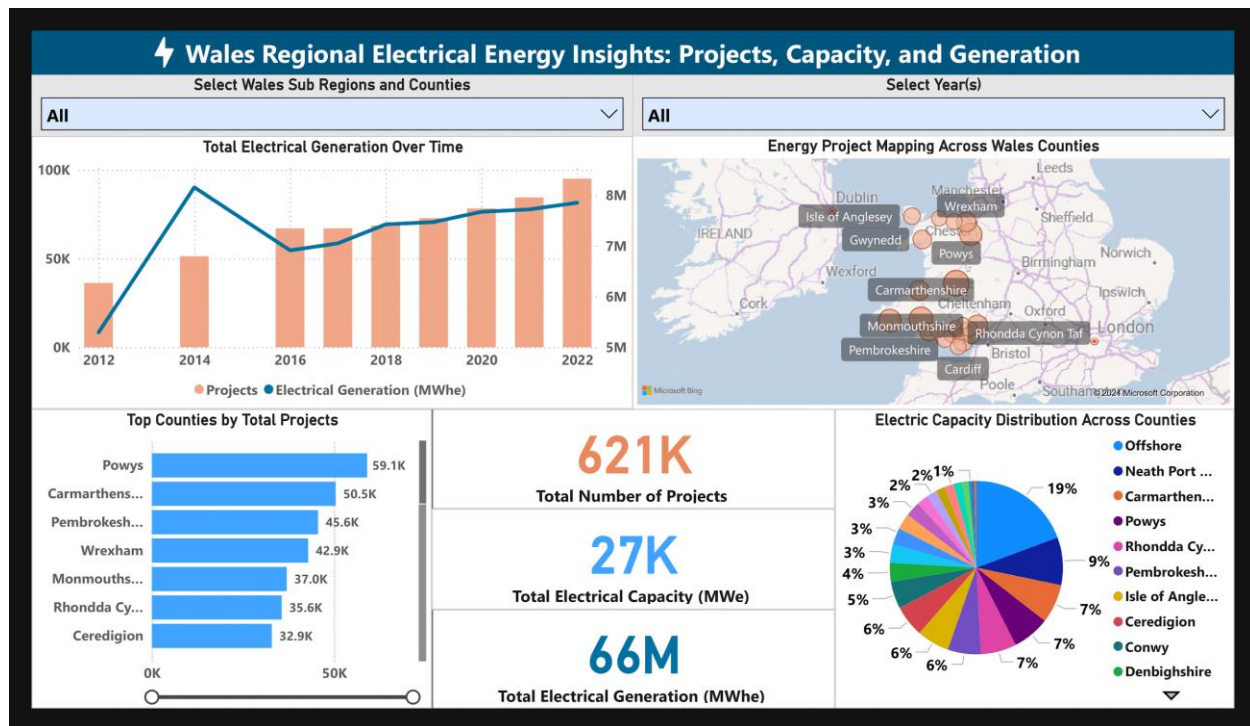


Figure 43: Dashboard for Wales Region

Total Electrical Generation Over Time: The growth in generation of electricity in MWhe and the number of projects is depicted from 2012 to 2022 in the form of line and bar graph in top left part of the figure. Though the project numbers have been unstable over the period, the electrical generation has been steadily increasing, touching approximately 8 million MWhe in 2022.

Energy Project Mapping Across Wales Counties: There is a map on the top right corner indicating distribution of energy projects with respect to the counties of Wales where POWYS, WREXHAM, Carmarthenshire and PEMBROKESHIRE are highlighted as energy zones. That can be filtered using the Filter given above chart.

Top Counties by Total Projects: Below left side graph, there is a bar chart showing the leading counties that have witnessed implementation of most energy projects. Powys tops the chart with 59,100 projects, followed by Carmarthenshire with 50,500 projects and Pembrokeshire with 45,600 projects.

Total Numbers: The number of measures considered pertinent for extensive depiction is less and these are highlighted on the dashboard:

- 621K Total number of projects for all regional classifications.
- 27K MWe of the total electrical capacity.

- 66M MWhe of total electrical generation suggesting the big picture of energy generation.

Electric Capacity Distribution: The pie chart over the bottom right indicates the share distribution of electrical capacity across the various counties; Offshore takes 19%; Neath Port Talbot, 9%; Carmarthenshire, 7%; and Powys, 7%. Other regions add much smaller shares to the total energy capacity.

To sum up, based on the number of projects, their distribution, capacity, and generation in certain years, this dashboard, considering the average and total generation, represents Wales' energy situation.

Similarly, the visualization results for GREP chart

- Total Hydro Energy: 258K units of hydro energy shown by blue icon.
- Total Solar Energy: 127K units of solar energy depicted by a yellow sun.
- Total Wind Energy: 206K units of wind energy with the pinpoint denoted by a pink wind icon.

These figures are a total of each class of renewable energy in the global market.

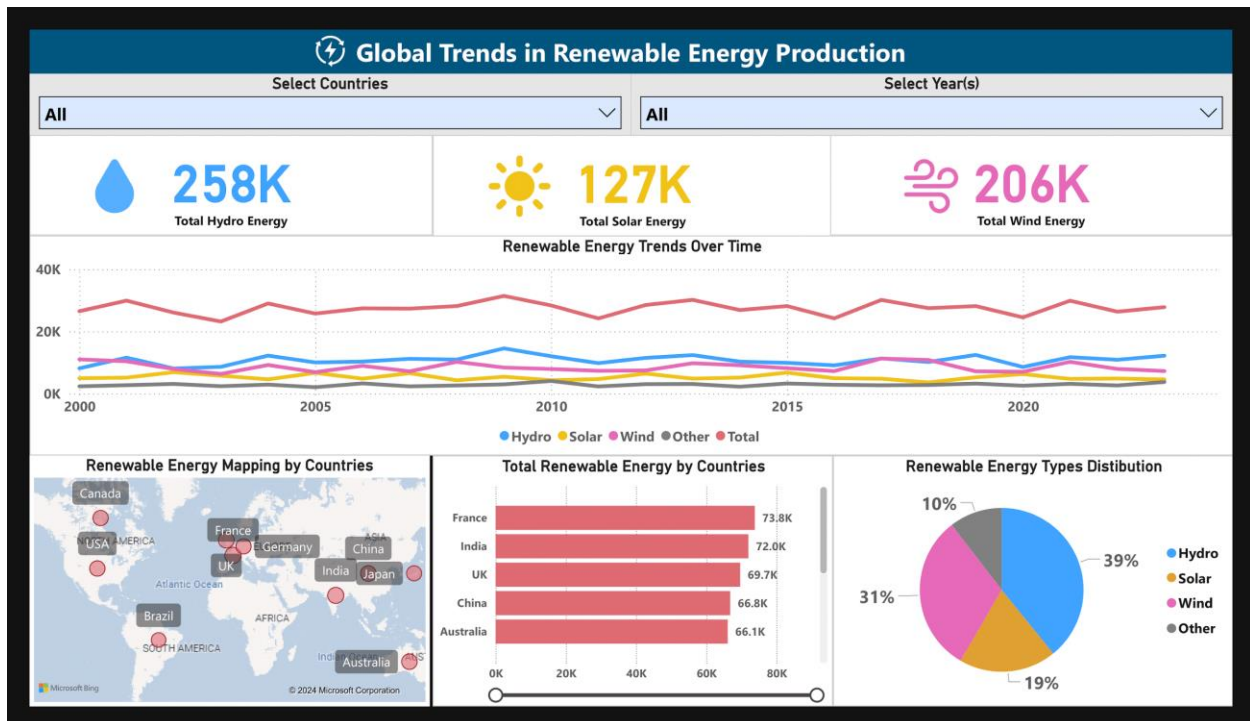


Figure 44: Dashboard for GREP

In addition to overall sum, line chart tracks global renewable energy production from 2000 to 2020 across various types:

- Hydro: Gradual increase over time.
- Solar: Relatively stable and fairly steady after 2015.
- Wind: It rose sharply after the middle of 2000s there first and foremost.
- Total: Total averages energy trend describing constant raise with tiny oscillations. This visualization shows the gradual increase in renewable energy generation, but especially in wind energy, and more gradually in solar energy.

Dashboard further adds list of top 5 countries, providing highest percentages of renewable energy. Given in bottom center chart

- France: 73.8K units.
- India: 72.0K units.
- UK: 69.7K units.
- China: 66.8K units.
- Australia: 66.1K units.

These are member countries in the renewable energy international market, with France ranked top among the countries.

Finally, the pie chart provides the generation of renewable power by country across the world. Hydroelectric power dominates the world's renewable energy output in terms of capacity representing 39% of total renewable generated power. Wind energy accounts for the second largest part of 31% while solar power comes at 19% of the total. Of these three, hydroelectric, wind, and solar are by far the largest sources of renewable power being produced on a global scale. While hydropower stands as the clean energy source with the highest percentage, the percentage of solar power is gradually rising over a given time, an indication of its rising importance in the provision of clean energy in the world.

5. Conclusion:

The main purpose of this research was to use the real time State Wales data and used trend analysis, and data-driven approaches to forecast the electrical capacity, and electrical generation in the region using different time series models. In this way, detailed step by step methodology employed and created different visualization, trend analysis, correlation matrix and importantly development of dashboard to find the trends and patterns inside the large data helpful policy makers to create easiness in decision making.

The findings showed that the renewable energy in Wales is uneven across regions, and some technologies such as Anaerobic Digestion and Biomass Electricity experienced large fluctuations in the installed capacity, while others, such as Hydropower and Onshore Wind experienced much smaller changes. Applying correlation analysis, it was established that heat and electrical capacities have significantly inversely correlation suggesting a tradeoff. For the Wales data forecasted electrical generation and capacity showing declining trend till 2027. When it comes to electricity production from renewable sources globally, the general trend has been relatively flat – with solar slowly increasing, wind has remained mostly static and hydro electricity production increasing with minor fluctuations. However, the total renewable energy production in terms of GREP data did not show tremendous variation over the forecasted years. The analysis of the relationship between the renewable energy data of Wales and the available data from around the world yielded almost no connection between the energy capacities of Wales and the worldwide production of renewable energy. This means that the Welsh power from renewable sources is not significantly dictated by the conditions in the rest of the world.

In addition to the obtained results, research objectives were also successfully met. For instance, two big datasets collected from different sources containing diverse columns and different energy types were gathered and preprocessed to forecast the growing demand of energy in Wales and around the world. The datasets offered considerable variance on the total electrical and heat capacities by technologies, such as wind, solar, hydropower, anaerobic digestion, and biomass electricity, all in energy production. Forecasting results using the ARIMA model show that Wales is going down in the next 5 years for electrical capacity and electrical generation. The fluctuations in energy sources in Wales does not directly link to the word renewable energy. Finally, this research contains all hidden insights, of the operation of Wales' power sector along with the possible alterations and their effects within the future.

5.1. Limitations:

The present research has limitations for instance, by using the historical statistics as the main tools of calculation, it becomes almost impossible to make forecasts which are adjusted for new externalities such as policy or technological changes that can lead to drastically different energy demands and supplies. Further, features in the data sets which are not consistent or there may be missing data for some regions which may not have considerable amounts of developed Renewable energy resources. Same like the cross-sectional analyses, correlation method also cannot establish causation because of its weak correlation, an indication that the local contexts of Wales do not always replicate the global process. Secondly, the fundamental assumption of the ARIMA model, such that the future will be similar to the past, is unlikely to be relevant to the evolving character of the energy industry.

5.2. Future Work:

Further work aims to obtain a larger dataset containing data of various energy types including non-renewable to better understand the energy picture of Wales. Also, when current data is integrated into the forecasting model for energy demand prediction using sophisticated algorithms, the accuracy of energy demand forecast would be higher. The analysis of changes in energy usage trends will also be beneficial for the study of changes in policies and technologies affecting the sector.

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